Chapter 7

Interprocessor Communication (IPC)

The Interprocessor Communications (IPC) module allows communication between the two CPU subsystems.

7.1 Introduction

This section details the IPC features that each CPU can use to request and share information. The IPC features are:

- Message RAMs
- IPC flags and interrupts
- IPC command registers
- Flash pump semaphore
- Clock configuration semaphore
- Free-running counter

All IPC features are independent of each other, and most do not require any specific data format.

There are also two registers for boot mode and status communication. Please refer to the boot ROM chapter for more information on these registers.

Figure 7-1 shows the design structure of the IPC module.

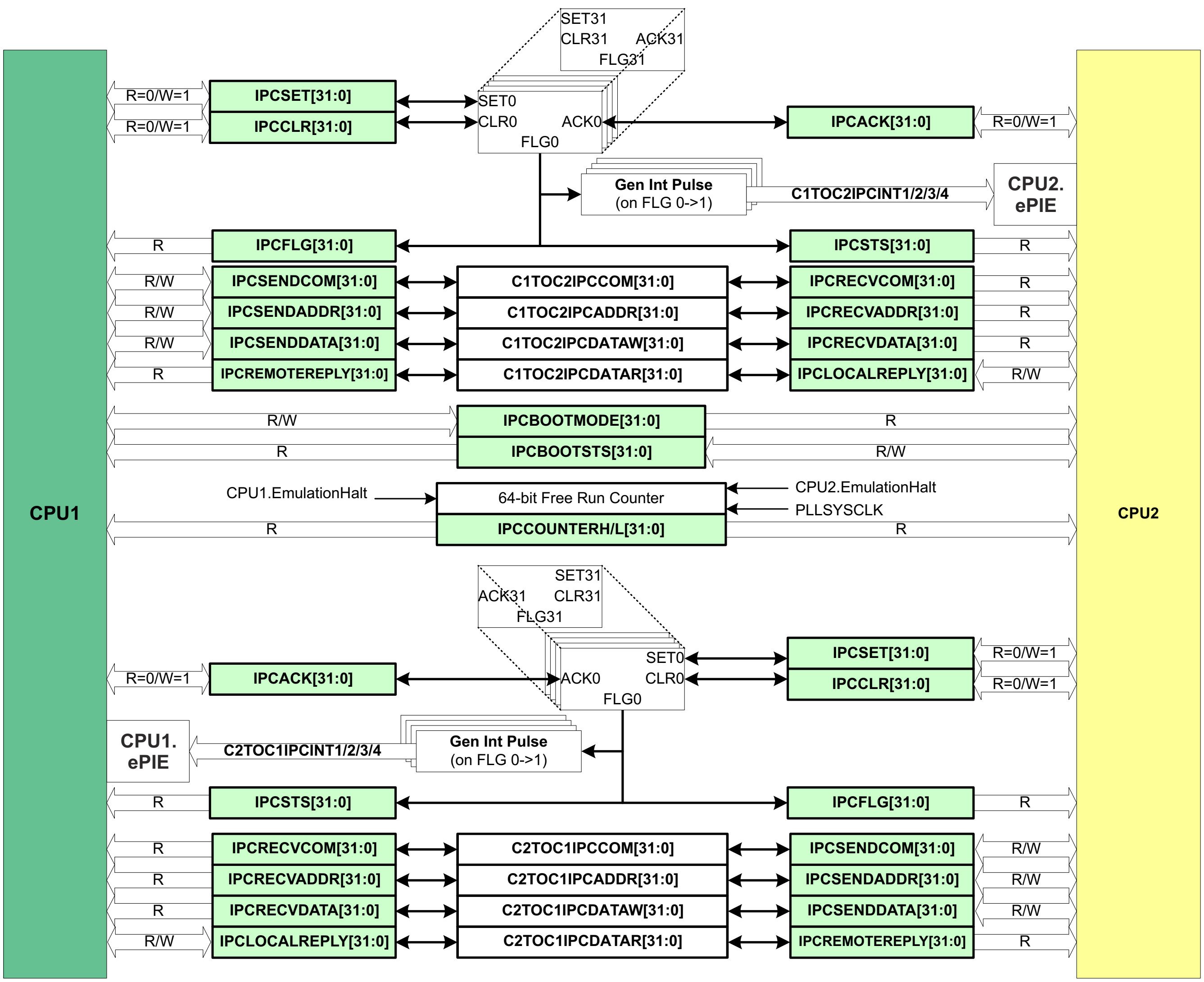

Figure 7-1. IPC Module Architecture

7.2 Message RAMs

There are two dedicated 2-kB blocks of message RAM. Each CPU and the DMA have read and write access to one RAM and read-only access to the other RAM, as shown in Table 7-1..

Reading or writing a message RAM does not trigger any events on the remote CPU.

Table 7-1. IPC Message RAM Read/Write Access

	CPU1	CPU2	CPU1 DMA	CPU2 DMA
CPU1 to CPU2 (1K x 16, address 0x03FC00)	R/W	R	R/W	R
CPU2 to CPU1 (1K x 16, address 0x03F800)	R	R/W	R	R/W

7.3 IPC Flags and Interrupts

There are 32 IPC event signals in each direction between the CPU pairs. These signals can be used for flag-based event polling. With the C28x core, four of them (IPC0 - IPC3) can be configured to generate IPC interrupts on the remote CPU.

7.4 IPC Command Registers

The IPC command registers provide a simple and flexible way for the CPUs to exchange more complex messages. Each CPU has eight dedicated registers; four for sending messages and four for receiving messages. The register names were chosen to support a simple command/response protocol, but can be used for any purpose. Only the read/write permissions are determined by hardware; the data format is entirely software-defined.

For sending messages, each CPU has three writable registers and one read-only register. Those same registers are accessible on the remote CPU as three read-only registers and one writable register. Table 7-2 shows the command registers.

Table 7-2. IPC Command Registers

Local Register Name	Local CPU	Remote CPU	Remote Register Name
IPCSENDCOM	R/W	R	IPCRECVCOM
IPCSENDADDR	R/W	R	IPCRECVADDR
IPCSENDDATA	R/W	R	IPCRECVDATA
IPCREMOTEREPLY	R	R/W	IPCLOCALREPLY

7.5 Free-Running Counter

A 64-bit free-running counter is present in the device and can be used to timestamp IPC events between processors. The counter is clocked by PLLSYSCLK and reset by SYSRSn. The counter is implemented as two 32-bit registers, IPCCOUNTERH and IPCCOUNTERL. When IPCCOUNTERL is read, the value of IPCCOUNTERH is saved. A subsequent read to IPCCOUNTERH returns this saved value. Therefore, the user must always read IPCCOUNTERL first then read IPCCOUNTERH next. This design prevents race conditions due to IPCCOUNTERL overflowing between reads of the two registers.

The free-running counter stops only when emulation is suspended (when debugger hits a breakpoint) on all CPUs. If any core is executing, the counter runs.

7.6 IPC Communication Protocol

This section describes the hardware support options for IPC communication between the two CPUs. These options can be used independently or in combination. All flag definitions and data formats are entirely user-defined.

- The flag system supports event-based communication via interrupts and register polling.
  - CPUx can raise an IPC event by writing to any of the 32 bits of the IPCSET register. This sets the corresponding bits in the CPUx IPCFLG register and CPUy IPCSTS register.
  - CPUy can signal the response to the event by setting the appropriate bit in the IPCACK register. This clears the corresponding bits in the CPUx IPCFLG register and the CPUy IPCSTS register.
  - If CPUx needs to cancel an event, CPUx can set the appropriate bit in the IPCCLR register. This has the same effect as CPUy writing to IPCACK.
  - Flags 0–3 (set using IPCSET[3:0]) fire interrupts to the remote CPU. The remote CPU must configure the ePIE module properly to receive an IPC interrupt. Flags 4–31 (set using IPCSET[31:4]) do not produce interrupts. Multiple flags can be set, acknowledged, and cleared simultaneously.
- The command registers support sending several distinct pieces of information and are named COM, ADDR, DATA, and REPLY for convenience only and can hold whatever data the application needs.
  - CPUx can write data to the IPCSENDCOM, IPCSENDADDR, and IPCSENDDATA registers. CPUy receives these in the IPCRECVCOM, IPCRECVADDR, and IPCRECVDATA registers.
  - CPUy can respond by writing to its IPCLOCALREPLY register. CPUx receives this data in its ownIPCREMOTEREPLY register.
- There is an additional pair of command-like registers offered for boot-time IPC or any other convenient use — IPCBOOTMODE and IPCBOOTSTS. Both CPUs can read these registers. CPUx can only write to IPCBOOTMODE, and CPUy can only write to IPCBOOTSTS.
- There are two shared memories for passing large amounts of data between the CPUs. Each CPU has a writable memory for sending data and a read-only memory for receiving data.
- Here is an example of how to use these features together. CPUx needs some data from CPUy's LS RAM. The data is at CPUy address 0x9400 and is 0x80 16-bit words long. The protocol can be implemented like this:
  - CPUx writes 0x1 to IPCSENDCOM, defined in software to mean "copy data from address". CPUx writes the address (0x9400) to IPCSENDADDR and the data length (0x80) to IPCSENDDATA.
  - CPUx writes to IPCSET[3] and IPCSET[16]. Here, IPC flag 3 is configured to send an interrupt and IPCSET[16] is defined in software to indicate an incoming command. CPUx begins polling for IPCFLG[3] to go low.
  - CPUy receives the interrupt. In the interrupt handler, CPUy checks IPCSTS, finds that flag 16 is set, and runs a command processor.
  - CPUy reads the command (0x1) from IPCRECVCOM, the address (0x9400) from IPCRECVADDR, and the data length (0x80) from IPCRECVDATA. CPUy then copies the LS RAM data to an empty space in the writable shared memory starting at offset 0x210.
  - CPUy writes the shared memory address (0x210) to the IPCLOCALREPLY register. CPUy then writes to IPCACK[16] and IPCACK[3] to clear the flags and indicate completion of the command. CPUy's work is done.
  - CPUx sees IPCFLG[3] go low. CPUx reads IPCREMOTEREPLY to get the shared memory offset of the copied data (0x210).

7.7 IPC Registers

This section describes the Interprocessor Communication Registers.

7.7.1 IPC Base Addresses

Table 7-3. IPC Base Addresses

Device Registers	Register Name	Start Address	End Address
IpcRegs (CPU1)	IPC_REGS_CPU1	0x0005_0000	0x0005_0023
IpcRegs (CPU2)	IPC_REGS_CPU2	0x0005_0000	0x0005_0023

7.7.2 IPC_REGS_CPU1 Registers

Table 7-4 lists the memory-mapped registers for the IPC_REGS_CPU1 registers. All register offset addresses not listed in Table 7-4 should be considered as reserved locations and the register contents should not be modified.

Table 7-4. IPC_REGS_CPU1 Registers

Offset	Acronym	Register Name	Write Protection	Section
0h	IPCACK	IPC incoming flag clear (acknowledge) register		Go
2h	IPCSTS	IPC incoming flag status register		Go
4h	IPCSET	IPC remote flag set register		Go
6h	IPCCLR	IPC remote flag clear register		Go
8h	IPCFLG	IPC remote flag status register		Go
Ch	IPCCOUNTERL	IPC Counter Low Register		Go
Eh	IPCCOUNTERH	IPC Counter High Register		Go
10h	IPCSENDCOM	Local to Remote IPC Command Register		Go
12h	IPCSENDADDR	Local to Remote IPC Address Register		Go
14h	IPCSENDDATA	Local to Remote IPC Data Register		Go
16h	IPCREMOTEREPLY	Remote to Local IPC Reply Data Register		Go
18h	IPCRECVCOM	Remote to Local IPC Command Register		Go
1Ah	IPCRECVADDR	Remote to Local IPC Address Register		Go
1Ch	IPCRECVDATA	Remote to Local IPC Data Register		Go
1Eh	IPCLOCALREPLY	Local to Remote IPC Reply Data Register		Go
20h	IPCBOOTSTS	CPU2 to CPU1 IPC Boot Status Register		Go
22h	IPCBOOTMODE	CPU1 to CPU2 IPC Boot Mode Register		Go

Complex bit access types are encoded to fit into small table cells. Table 7-5 shows the codes that are used for access types in this section.

Table 7-5. IPC_REGS_CPU1 Access Type Codes

Access Type	Code	Description
Read Type		
R	R	Read
R-0	R -0	Read Returns 0s
Write Type		
W	W	Write
W1S	W 1S	Write 1 to set
Reset or Default Value		
-n		Value after reset or the default value
Register Array Variables		
i,j,k,l,m,n		When these variables are used in a register name, an offset, or an address, they refer to the value of a register array where the register is part of a group of repeating registers. The register groups form a hierarchical structure and the array is represented with a formula.
y		When this variable is used in a register name, an offset, or an address it refers to the value of a register array.

7.7.2.1 IPCACK Register (Offset = 0h) [Reset = 00000000h]

IPCACK is shown in Figure 7-2 and described in Table 7-6.

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IPC incoming flag clear (acknowledge) register

Figure 7-2. IPCACK Register

31	30	29	28	27	26	25	24
IPC31	IPC30	IPC29	IPC28	IPC27	IPC26	IPC25	IPC24
R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h
23	22	21	20	19	18	17	16
IPC23	IPC22	IPC21	IPC20	IPC19	IPC18	IPC17	IPC16
R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h
15	14	13	12	11	10	9	8
IPC15	IPC14	IPC13	IPC12	IPC11	IPC10	IPC9	IPC8
R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h
7	6	5	4	3	2	1	0
IPC7	IPC6	IPC5	IPC4	IPC3	IPC2	IPC1	IPC0
R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h

Table 7-6. IPCACK Register Field Descriptions

Bit	Field	Type	Reset	Description
31	IPC31	R-0/W1S	0h	Writing 1 to this bit clears the IPC31 event flag which was set by the remote CPU. Writing 0 to this bit has no effect. Reset type: SYSRSn
30	IPC30	R-0/W1S	0h	Writing 1 to this bit clears the IPC30 event flag which was set by the remote CPU. Writing 0 to this bit has no effect. Reset type: SYSRSn
29	IPC29	R-0/W1S	0h	Writing 1 to this bit clears the IPC29 event flag which was set by the remote CPU. Writing 0 to this bit has no effect. Reset type: SYSRSn
28	IPC28	R-0/W1S	0h	Writing 1 to this bit clears the IPC28 event flag which was set by the remote CPU. Writing 0 to this bit has no effect. Reset type: SYSRSn
27	IPC27	R-0/W1S	0h	Writing 1 to this bit clears the IPC27 event flag which was set by the remote CPU. Writing 0 to this bit has no effect. Reset type: SYSRSn
26	IPC26	R-0/W1S	0h	Writing 1 to this bit clears the IPC26 event flag which was set by the remote CPU. Writing 0 to this bit has no effect. Reset type: SYSRSn
25	IPC25	R-0/W1S	0h	Writing 1 to this bit clears the IPC25 event flag which was set by the remote CPU. Writing 0 to this bit has no effect. Reset type: SYSRSn

Table 7-6. IPCACK Register Field Descriptions (continued)

Bit	Field	Type	Reset	Description
24	IPC24	R-0/W1S	0h	Writing 1 to this bit clears the IPC24 event flag which was set by the remote CPU. Writing 0 to this bit has no effect. Reset type: SYSRSn
23	IPC23	R-0/W1S	0h	Writing 1 to this bit clears the IPC23 event flag which was set by the remote CPU. Writing 0 to this bit has no effect. Reset type: SYSRSn
22	IPC22	R-0/W1S	0h	Writing 1 to this bit clears the IPC22 event flag which was set by the remote CPU. Writing 0 to this bit has no effect. Reset type: SYSRSn
21	IPC21	R-0/W1S	0h	Writing 1 to this bit clears the IPC21 event flag which was set by the remote CPU. Writing 0 to this bit has no effect. Reset type: SYSRSn
20	IPC20	R-0/W1S	0h	Writing 1 to this bit clears the IPC20 event flag which was set by the remote CPU. Writing 0 to this bit has no effect. Reset type: SYSRSn
19	IPC19	R-0/W1S	0h	Writing 1 to this bit clears the IPC19 event flag which was set by the remote CPU. Writing 0 to this bit has no effect. Reset type: SYSRSn
18	IPC18	R-0/W1S	0h	Writing 1 to this bit clears the IPC18 event flag which was set by the remote CPU. Writing 0 to this bit has no effect. Reset type: SYSRSn
17	IPC17	R-0/W1S	0h	Writing 1 to this bit clears the IPC17 event flag which was set by the remote CPU. Writing 0 to this bit has no effect. Reset type: SYSRSn
16	IPC16	R-0/W1S	0h	Writing 1 to this bit clears the IPC16 event flag which was set by the remote CPU. Writing 0 to this bit has no effect. Reset type: SYSRSn
15	IPC15	R-0/W1S	0h	Writing 1 to this bit clears the IPC15 event flag which was set by the remote CPU. Writing 0 to this bit has no effect. Reset type: SYSRSn
14	IPC14	R-0/W1S	0h	Writing 1 to this bit clears the IPC14 event flag which was set by the remote CPU. Writing 0 to this bit has no effect. Reset type: SYSRSn
13	IPC13	R-0/W1S	0h	Writing 1 to this bit clears the IPC13 event flag which was set by the remote CPU. Writing 0 to this bit has no effect. Reset type: SYSRSn
12	IPC12	R-0/W1S	0h	Writing 1 to this bit clears the IPC12 event flag which was set by the remote CPU. Writing 0 to this bit has no effect. Reset type: SYSRSn
11	IPC11	R-0/W1S	0h	Writing 1 to this bit clears the IPC11 event flag which was set by the remote CPU. Writing 0 to this bit has no effect. Reset type: SYSRSn

Table 7-6. IPCACK Register Field Descriptions (continued)

Bit	Field	Type	Reset	Description
10	IPC10	R-0/W1S	0h	Writing 1 to this bit clears the IPC10 event flag which was set by the remote CPU. Writing 0 to this bit has no effect. Reset type: SYSRSn
9	IPC9	R-0/W1S	0h	Writing 1 to this bit clears the IPC9 event flag which was set by the remote CPU. Writing 0 to this bit has no effect. Reset type: SYSRSn
8	IPC8	R-0/W1S	0h	Writing 1 to this bit clears the IPC8 event flag which was set by the remote CPU. Writing 0 to this bit has no effect. Reset type: SYSRSn
7	IPC7	R-0/W1S	0h	Writing 1 to this bit clears the IPC7 event flag which was set by the remote CPU. Writing 0 to this bit has no effect. Reset type: SYSRSn
6	IPC6	R-0/W1S	0h	Writing 1 to this bit clears the IPC6 event flag which was set by the remote CPU. Writing 0 to this bit has no effect. Reset type: SYSRSn
5	IPC5	R-0/W1S	0h	Writing 1 to this bit clears the IPC5 event flag which was set by the remote CPU. Writing 0 to this bit has no effect. Reset type: SYSRSn
4	IPC4	R-0/W1S	0h	Writing 1 to this bit clears the IPC4 event flag which was set by the remote CPU. Writing 0 to this bit has no effect. Reset type: SYSRSn
3	IPC3	R-0/W1S	0h	Writing 1 to this bit clears the IPC3 event flag which was set by the remote CPU. Writing 0 to this bit has no effect. Reset type: SYSRSn
2	IPC2	R-0/W1S	0h	Writing 1 to this bit clears the IPC2 event flag which was set by the remote CPU. Writing 0 to this bit has no effect. Reset type: SYSRSn
1	IPC1	R-0/W1S	0h	Writing 1 to this bit clears the IPC1 event flag which was set by the remote CPU. Writing 0 to this bit has no effect. Reset type: SYSRSn
0	IPC0	R-0/W1S	0h	Writing 1 to this bit clears the IPC0 event flag which was set by the remote CPU. Writing 0 to this bit has no effect. Reset type: SYSRSn

7.7.2.2 IPCSTS Register (Offset = 2h) [Reset = 00000000h]

IPCSTS is shown in Figure 7-3 and described in Table 7-7.

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IPC incoming flag status register

Figure 7-3. IPCSTS Register

31	30	29	28	27	26	25	24
IPC31	IPC30	IPC29	IPC28	IPC27	IPC26	IPC25	IPC24
R-0h	R-0h	R-0h	R-0h	R-0h	R-0h	R-0h	R-0h
23	22	21	20	19	18	17	16
IPC23	IPC22	IPC21	IPC20	IPC19	IPC18	IPC17	IPC16
R-0h	R-0h	R-0h	R-0h	R-0h	R-0h	R-0h	R-0h
15	14	13	12	11	10	9	8
IPC15	IPC14	IPC13	IPC12	IPC11	IPC10	IPC9	IPC8
R-0h	R-0h	R-0h	R-0h	R-0h	R-0h	R-0h	R-0h
7	6	5	4	3	2	1	0
IPC7	IPC6	IPC5	IPC4	IPC3	IPC2	IPC1	IPC0
R-0h	R-0h	R-0h	R-0h	R-0h	R-0h	R-0h	R-0h

Table 7-7. IPCSTS Register Field Descriptions

Bit	Field	Type	Reset	Description
31	IPC31	R	0h	Indicates to the local CPU if the IPC31 event flag was set by the remote CPU. 0: No IPC31 event was set by the remote CPU 1: An IPC31 event was set by the remote CPU Reset type: SYSRSn
30	IPC30	R	0h	Indicates to the local CPU if the IPC30 event flag was set by the remote CPU. 0: No IPC30 event was set by the remote CPU 1: An IPC30 event was set by the remote CPU Reset type: SYSRSn
29	IPC29	R	0h	Indicates to the local CPU if the IPC29 event flag was set by the remote CPU. 0: No IPC29 event was set by the remote CPU 1: An IPC29 event was set by the remote CPU Reset type: SYSRSn
28	IPC28	R	0h	Indicates to the local CPU if the IPC28 event flag was set by the remote CPU. 0: No IPC28 event was set by the remote CPU 1: An IPC28 event was set by the remote CPU Reset type: SYSRSn
27	IPC27	R	0h	Indicates to the local CPU if the IPC27 event flag was set by the remote CPU. 0: No IPC27 event was set by the remote CPU 1: An IPC27 event was set by the remote CPU Reset type: SYSRSn
26	IPC26	R	0h	Indicates to the local CPU if the IPC26 event flag was set by the remote CPU. 0: No IPC26 event was set by the remote CPU 1: An IPC26 event was set by the remote CPU Reset type: SYSRSn

Table 7-7. IPCSTS Register Field Descriptions (continued)

Bit	Field	Type	Reset	Description
25	IPC25	R	0h	Indicates to the local CPU if the IPC25 event flag was set by the remote CPU. 0: No IPC25 event was set by the remote CPU 1: An IPC25 event was set by the remote CPU Reset type: SYSRSn
24	IPC24	R	0h	Indicates to the local CPU if the IPC24 event flag was set by the remote CPU. 0: No IPC24 event was set by the remote CPU 1: An IPC24 event was set by the remote CPU Reset type: SYSRSn
23	IPC23	R	0h	Indicates to the local CPU if the IPC23 event flag was set by the remote CPU. 0: No IPC23 event was set by the remote CPU 1: An IPC23 event was set by the remote CPU Reset type: SYSRSn
22	IPC22	R	0h	Indicates to the local CPU if the IPC22 event flag was set by the remote CPU. 0: No IPC22 event was set by the remote CPU 1: An IPC22 event was set by the remote CPU Reset type: SYSRSn
21	IPC21	R	0h	Indicates to the local CPU if the IPC21 event flag was set by the remote CPU. 0: No IPC21 event was set by the remote CPU 1: An IPC21 event was set by the remote CPU Reset type: SYSRSn
20	IPC20	R	0h	Indicates to the local CPU if the IPC20 event flag was set by the remote CPU. 0: No IPC20 event was set by the remote CPU 1: An IPC20 event was set by the remote CPU Reset type: SYSRSn
19	IPC19	R	0h	Indicates to the local CPU if the IPC19 event flag was set by the remote CPU. 0: No IPC19 event was set by the remote CPU 1: An IPC19 event was set by the remote CPU Reset type: SYSRSn
18	IPC18	R	0h	Indicates to the local CPU if the IPC18 event flag was set by the remote CPU. 0: No IPC18 event was set by the remote CPU 1: An IPC18 event was set by the remote CPU Reset type: SYSRSn
17	IPC17	R	0h	Indicates to the local CPU if the IPC17 event flag was set by the remote CPU. 0: No IPC17 event was set by the remote CPU 1: An IPC17 event was set by the remote CPU Reset type: SYSRSn
16	IPC16	R	0h	Indicates to the local CPU if the IPC16 event flag was set by the remote CPU. 0: No IPC16 event was set by the remote CPU 1: An IPC16 event was set by the remote CPU Reset type: SYSRSn
15	IPC15	R	0h	Indicates to the local CPU if the IPC15 event flag was set by the remote CPU. 0: No IPC15 event was set by the remote CPU 1: An IPC15 event was set by the remote CPU Reset type: SYSRSn

Table 7-7. IPCSTS Register Field Descriptions (continued)

Bit	Field	Type	Reset	Description
14	IPC14	R	0h	Indicates to the local CPU if the IPC14 event flag was set by the remote CPU. 0: No IPC14 event was set by the remote CPU 1: An IPC14 event was set by the remote CPU Reset type: SYSRSn
13	IPC13	R	0h	Indicates to the local CPU if the IPC13 event flag was set by the remote CPU. 0: No IPC13 event was set by the remote CPU 1: An IPC13 event was set by the remote CPU Reset type: SYSRSn
12	IPC12	R	0h	Indicates to the local CPU if the IPC12 event flag was set by the remote CPU. 0: No IPC12 event was set by the remote CPU 1: An IPC12 event was set by the remote CPU Reset type: SYSRSn
11	IPC11	R	0h	Indicates to the local CPU if the IPC11 event flag was set by the remote CPU. 0: No IPC11 event was set by the remote CPU 1: An IPC11 event was set by the remote CPU Reset type: SYSRSn
10	IPC10	R	0h	Indicates to the local CPU if the IPC10 event flag was set by the remote CPU. 0: No IPC10 event was set by the remote CPU 1: An IPC10 event was set by the remote CPU Reset type: SYSRSn
9	IPC9	R	0h	Indicates to the local CPU if the IPC9 event flag was set by the remote CPU. 0: No IPC9 event was set by the remote CPU 1: An IPC9 event was set by the remote CPU Reset type: SYSRSn
8	IPC8	R	0h	Indicates to the local CPU if the IPC8 event flag was set by the remote CPU. 0: No IPC8 event was set by the remote CPU 1: An IPC8 event was set by the remote CPU Reset type: SYSRSn
7	IPC7	R	0h	Indicates to the local CPU if the IPC7 event flag was set by the remote CPU. 0: No IPC7 event was set by the remote CPU 1: An IPC7 event was set by the remote CPU Reset type: SYSRSn
6	IPC6	R	0h	Indicates to the local CPU if the IPC6 event flag was set by the remote CPU. 0: No IPC6 event was set by the remote CPU 1: An IPC6 event was set by the remote CPU Reset type: SYSRSn
5	IPC5	R	0h	Indicates to the local CPU if the IPC5 event flag was set by the remote CPU. 0: No IPC5 event was set by the remote CPU 1: An IPC5 event was set by the remote CPU Reset type: SYSRSn
4	IPC4	R	0h	Indicates to the local CPU if the IPC4 event flag was set by the remote CPU. 0: No IPC4 event was set by the remote CPU 1: An IPC4 event was set by the remote CPU Reset type: SYSRSn

Table 7-7. IPCSTS Register Field Descriptions (continued)

Bit	Field	Type	Reset	Description
3	IPC3	R	0h	Indicates to the local CPU if the IPC3 event flag was set by the remote CPU. 0: No IPC3 event was set by the remote CPU 1: An IPC3 event was set by the remote CPU Notes [1] IPC event flags 0-3 will trigger interrupts in the receiving CPU via the ePIE. Reset type: SYSRSn
2	IPC2	R	0h	Indicates to the local CPU if the IPC2 event flag was set by the remote CPU. 0: No IPC2 event was set by the remote CPU 1: An IPC2 event was set by the remote CPU Notes [1] IPC event flags 0-3 will trigger interrupts in the receiving CPU via the ePIE. Reset type: SYSRSn
1	IPC1	R	0h	Indicates to the local CPU if the IPC1 event flag was set by the remote CPU. 0: No IPC1 event was set by the remote CPU 1: An IPC1 event was set by the remote CPU Notes [1] IPC event flags 0-3 will trigger interrupts in the receiving CPU via the ePIE. Reset type: SYSRSn
0	IPC0	R	0h	Indicates to the local CPU if the IPC0 event flag was set by the remote CPU. 0: No IPC0 event was set by the remote CPU 1: An IPC0 event was set by the remote CPU Notes [1] IPC event flags 0-3 will trigger interrupts in the receiving CPU via the ePIE. Reset type: SYSRSn

7.7.2.3 IPCSET Register (Offset = 4h) [Reset = 00000000h]

IPCSET is shown in Figure 7-4 and described in Table 7-8.

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IPC remote flag set register

Figure 7-4. IPCSET Register

31	30	29	28	27	26	25	24
IPC31	IPC30	IPC29	IPC28	IPC27	IPC26	IPC25	IPC24
R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h
23	22	21	20	19	18	17	16
IPC23	IPC22	IPC21	IPC20	IPC19	IPC18	IPC17	IPC16
R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h
15	14	13	12	11	10	9	8
IPC15	IPC14	IPC13	IPC12	IPC11	IPC10	IPC9	IPC8
R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h
7	6	5	4	3	2	1	0
IPC7	IPC6	IPC5	IPC4	IPC3	IPC2	IPC1	IPC0
R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h

Table 7-8. IPCSET Register Field Descriptions

Bit	Field	Type	Reset	Description
31	IPC31	R-0/W1S	0h	Writing 1 to this bit sets the IPC31 event flag for the remote CPU. Writing 0 has no effect. Reset type: SYSRSn
30	IPC30	R-0/W1S	0h	Writing 1 to this bit sets the IPC30 event flag for the remote CPU. Writing 0 has no effect. Reset type: SYSRSn
29	IPC29	R-0/W1S	0h	Writing 1 to this bit sets the IPC29 event flag for the remote CPU. Writing 0 has no effect. Reset type: SYSRSn
28	IPC28	R-0/W1S	0h	Writing 1 to this bit sets the IPC28 event flag for the remote CPU. Writing 0 has no effect. Reset type: SYSRSn
27	IPC27	R-0/W1S	0h	Writing 1 to this bit sets the IPC27 event flag for the remote CPU. Writing 0 has no effect. Reset type: SYSRSn
26	IPC26	R-0/W1S	0h	Writing 1 to this bit sets the IPC26 event flag for the remote CPU. Writing 0 has no effect. Reset type: SYSRSn
25	IPC25	R-0/W1S	0h	Writing 1 to this bit sets the IPC25 event flag for the remote CPU. Writing 0 has no effect. Reset type: SYSRSn
24	IPC24	R-0/W1S	0h	Writing 1 to this bit sets the IPC24 event flag for the remote CPU. Writing 0 has no effect. Reset type: SYSRSn
23	IPC23	R-0/W1S	0h	Writing 1 to this bit sets the IPC23 event flag for the remote CPU. Writing 0 has no effect. Reset type: SYSRSn
22	IPC22	R-0/W1S	0h	Writing 1 to this bit sets the IPC22 event flag for the remote CPU. Writing 0 has no effect. Reset type: SYSRSn

Table 7-8. IPCSET Register Field Descriptions (continued)

Bit	Field	Type	Reset	Description
21	IPC21	R-0/W1S	0h	Writing 1 to this bit sets the IPC21 event flag for the remote CPU. Writing 0 has no effect. Reset type: SYSRSn
20	IPC20	R-0/W1S	0h	Writing 1 to this bit sets the IPC20 event flag for the remote CPU. Writing 0 has no effect. Reset type: SYSRSn
19	IPC19	R-0/W1S	0h	Writing 1 to this bit sets the IPC19 event flag for the remote CPU. Writing 0 has no effect. Reset type: SYSRSn
18	IPC18	R-0/W1S	0h	Writing 1 to this bit sets the IPC18 event flag for the remote CPU. Writing 0 has no effect. Reset type: SYSRSn
17	IPC17	R-0/W1S	0h	Writing 1 to this bit sets the IPC17 event flag for the remote CPU. Writing 0 has no effect. Reset type: SYSRSn
16	IPC16	R-0/W1S	0h	Writing 1 to this bit sets the IPC16 event flag for the remote CPU. Writing 0 has no effect. Reset type: SYSRSn
15	IPC15	R-0/W1S	0h	Writing 1 to this bit sets the IPC15 event flag for the remote CPU. Writing 0 has no effect. Reset type: SYSRSn
14	IPC14	R-0/W1S	0h	Writing 1 to this bit sets the IPC14 event flag for the remote CPU. Writing 0 has no effect. Reset type: SYSRSn
13	IPC13	R-0/W1S	0h	Writing 1 to this bit sets the IPC13 event flag for the remote CPU. Writing 0 has no effect. Reset type: SYSRSn
12	IPC12	R-0/W1S	0h	Writing 1 to this bit sets the IPC12 event flag for the remote CPU. Writing 0 has no effect. Reset type: SYSRSn
11	IPC11	R-0/W1S	0h	Writing 1 to this bit sets the IPC11 event flag for the remote CPU. Writing 0 has no effect. Reset type: SYSRSn
10	IPC10	R-0/W1S	0h	Writing 1 to this bit sets the IPC10 event flag for the remote CPU. Writing 0 has no effect. Reset type: SYSRSn
9	IPC9	R-0/W1S	0h	Writing 1 to this bit sets the IPC9 event flag for the remote CPU. Writing 0 has no effect. Reset type: SYSRSn
8	IPC8	R-0/W1S	0h	Writing 1 to this bit sets the IPC8 event flag for the remote CPU. Writing 0 has no effect. Reset type: SYSRSn
7	IPC7	R-0/W1S	0h	Writing 1 to this bit sets the IPC7 event flag for the remote CPU. Writing 0 has no effect. Reset type: SYSRSn
6	IPC6	R-0/W1S	0h	Writing 1 to this bit sets the IPC6 event flag for the remote CPU. Writing 0 has no effect. Reset type: SYSRSn
5	IPC5	R-0/W1S	0h	Writing 1 to this bit sets the IPC5 event flag for the remote CPU. Writing 0 has no effect. Reset type: SYSRSn

Table 7-8. IPCSET Register Field Descriptions (continued)

Bit	Field	Type	Reset	Description
4	IPC4	R-0/W1S	0h	Writing 1 to this bit sets the IPC4 event flag for the remote CPU. Writing 0 has no effect. Reset type: SYSRSn
3	IPC3	R-0/W1S	0h	Writing 1 to this bit sets the IPC3 event flag for the remote CPU. Writing 0 has no effect. Notes: [1] IPC event flags 0-3 will trigger interrupts in the receiving CPU via the ePIE. Reset type: SYSRSn
2	IPC2	R-0/W1S	0h	Writing 1 to this bit sets the IPC2 event flag for the remote CPU. Writing 0 has no effect. Notes: [1] IPC event flags 0-3 will trigger interrupts in the receiving CPU via the ePIE. Reset type: SYSRSn
1	IPC1	R-0/W1S	0h	Writing 1 to this bit sets the IPC1 event flag for the remote CPU. Writing 0 has no effect. Notes: [1] IPC event flags 0-3 will trigger interrupts in the receiving CPU via the ePIE. Reset type: SYSRSn
0	IPC0	R-0/W1S	0h	Writing 1 to this bit sets the IPC0 event flag for the remote CPU. Writing 0 has no effect. Notes: [1] IPC event flags 0-3 will trigger interrupts in the receiving CPU via the ePIE. Reset type: SYSRSn

7.7.2.4 IPCCLR Register (Offset = 6h) [Reset = 00000000h]

IPCCLR is shown in Figure 7-5 and described in Table 7-9.

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IPC remote flag clear register

Figure 7-5. IPCCLR Register

31	30	29	28	27	26	25	24
IPC31	IPC30	IPC29	IPC28	IPC27	IPC26	IPC25	IPC24
R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h
23	22	21	20	19	18	17	16
IPC23	IPC22	IPC21	IPC20	IPC19	IPC18	IPC17	IPC16
R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h
15	14	13	12	11	10	9	8
IPC15	IPC14	IPC13	IPC12	IPC11	IPC10	IPC9	IPC8
R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h
7	6	5	4	3	2	1	0
IPC7	IPC6	IPC5	IPC4	IPC3	IPC2	IPC1	IPC0
R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h

Table 7-9. IPCCLR Register Field Descriptions

Bit	Field	Type	Reset	Description
31	IPC31	R-0/W1S	0h	Writing 1 to this bit clears the IPC31 flag for the remote CPU. Writing 0 has no effect. Notes: [1] Normally, each CPU will clear (acknowledge) only its own local flags. This mechanism may be useful if the remote CPU is non-responsive. Reset type: SYSRSn
30	IPC30	R-0/W1S	0h	Writing 1 to this bit clears the IPC30 flag for the remote CPU. Writing 0 has no effect. Notes: [1] Normally, each CPU will clear (acknowledge) only its own local flags. This mechanism may be useful if the remote CPU is non-responsive. Reset type: SYSRSn
29	IPC29	R-0/W1S	0h	Writing 1 to this bit clears the IPC29 flag for the remote CPU. Writing 0 has no effect. Notes: [1] Normally, each CPU will clear (acknowledge) only its own local flags. This mechanism may be useful if the remote CPU is non-responsive. Reset type: SYSRSn
28	IPC28	R-0/W1S	0h	Writing 1 to this bit clears the IPC28 flag for the remote CPU. Writing 0 has no effect. Notes: [1] Normally, each CPU will clear (acknowledge) only its own local flags. This mechanism may be useful if the remote CPU is non-responsive. Reset type: SYSRSn

Table 7-9. IPCCLR Register Field Descriptions (continued)

Bit	Field	Type	Reset	Description
27	IPC27	R-0/W1S	0h	Writing 1 to this bit clears the IPC27 flag for the remote CPU. Writing 0 has no effect. Notes: [1] Normally, each CPU will clear (acknowledge) only its own local flags. This mechanism may be useful if the remote CPU is non-responsive. Reset type: SYSRSn
26	IPC26	R-0/W1S	0h	Writing 1 to this bit clears the IPC26 flag for the remote CPU. Writing 0 has no effect. Notes: [1] Normally, each CPU will clear (acknowledge) only its own local flags. This mechanism may be useful if the remote CPU is non-responsive. Reset type: SYSRSn
25	IPC25	R-0/W1S	0h	Writing 1 to this bit clears the IPC25 flag for the remote CPU. Writing 0 has no effect. Notes: [1] Normally, each CPU will clear (acknowledge) only its own local flags. This mechanism may be useful if the remote CPU is non-responsive. Reset type: SYSRSn
24	IPC24	R-0/W1S	0h	Writing 1 to this bit clears the IPC24 flag for the remote CPU. Writing 0 has no effect. Notes: [1] Normally, each CPU will clear (acknowledge) only its own local flags. This mechanism may be useful if the remote CPU is non-responsive. Reset type: SYSRSn
23	IPC23	R-0/W1S	0h	Writing 1 to this bit clears the IPC23 flag for the remote CPU. Writing 0 has no effect. Notes: [1] Normally, each CPU will clear (acknowledge) only its own local flags. This mechanism may be useful if the remote CPU is non-responsive. Reset type: SYSRSn
22	IPC22	R-0/W1S	0h	Writing 1 to this bit clears the IPC22 flag for the remote CPU. Writing 0 has no effect. Notes: [1] Normally, each CPU will clear (acknowledge) only its own local flags. This mechanism may be useful if the remote CPU is non-responsive. Reset type: SYSRSn
21	IPC21	R-0/W1S	0h	Writing 1 to this bit clears the IPC21 flag for the remote CPU. Writing 0 has no effect. Notes: [1] Normally, each CPU will clear (acknowledge) only its own local flags. This mechanism may be useful if the remote CPU is non-responsive. Reset type: SYSRSn
20	IPC20	R-0/W1S	0h	Writing 1 to this bit clears the IPC20 flag for the remote CPU. Writing 0 has no effect. Notes: [1] Normally, each CPU will clear (acknowledge) only its own local flags. This mechanism may be useful if the remote CPU is non-responsive. Reset type: SYSRSn

Table 7-9. IPCCLR Register Field Descriptions (continued)

Bit	Field	Type	Reset	Description
19	IPC19	R-0/W1S	0h	Writing 1 to this bit clears the IPC19 flag for the remote CPU. Writing 0 has no effect. Notes: [1] Normally, each CPU will clear (acknowledge) only its own local flags. This mechanism may be useful if the remote CPU is non-responsive. Reset type: SYSRSn
18	IPC18	R-0/W1S	0h	Writing 1 to this bit clears the IPC18 flag for the remote CPU. Writing 0 has no effect. Notes: [1] Normally, each CPU will clear (acknowledge) only its own local flags. This mechanism may be useful if the remote CPU is non-responsive. Reset type: SYSRSn
17	IPC17	R-0/W1S	0h	Writing 1 to this bit clears the IPC17 flag for the remote CPU. Writing 0 has no effect. Notes: [1] Normally, each CPU will clear (acknowledge) only its own local flags. This mechanism may be useful if the remote CPU is non-responsive. Reset type: SYSRSn
16	IPC16	R-0/W1S	0h	Writing 1 to this bit clears the IPC16 flag for the remote CPU. Writing 0 has no effect. Notes: [1] Normally, each CPU will clear (acknowledge) only its own local flags. This mechanism may be useful if the remote CPU is non-responsive. Reset type: SYSRSn
15	IPC15	R-0/W1S	0h	Writing 1 to this bit clears the IPC15 flag for the remote CPU. Writing 0 has no effect. Notes: [1] Normally, each CPU will clear (acknowledge) only its own local flags. This mechanism may be useful if the remote CPU is non-responsive. Reset type: SYSRSn
14	IPC14	R-0/W1S	0h	Writing 1 to this bit clears the IPC14 flag for the remote CPU. Writing 0 has no effect. Notes: [1] Normally, each CPU will clear (acknowledge) only its own local flags. This mechanism may be useful if the remote CPU is non-responsive. Reset type: SYSRSn
13	IPC13	R-0/W1S	0h	Writing 1 to this bit clears the IPC13 flag for the remote CPU. Writing 0 has no effect. Notes: [1] Normally, each CPU will clear (acknowledge) only its own local flags. This mechanism may be useful if the remote CPU is non-responsive. Reset type: SYSRSn
12	IPC12	R-0/W1S	0h	Writing 1 to this bit clears the IPC12 flag for the remote CPU. Writing 0 has no effect. Notes: [1] Normally, each CPU will clear (acknowledge) only its own local flags. This mechanism may be useful if the remote CPU is non-responsive. Reset type: SYSRSn

Table 7-9. IPCCLR Register Field Descriptions (continued)

Bit	Field	Type	Reset	Description
11	IPC11	R-0/W1S	0h	Writing 1 to this bit clears the IPC11 flag for the remote CPU. Writing 0 has no effect. Notes: [1] Normally, each CPU will clear (acknowledge) only its own local flags. This mechanism may be useful if the remote CPU is non-responsive. Reset type: SYSRSn
10	IPC10	R-0/W1S	0h	Writing 1 to this bit clears the IPC10 flag for the remote CPU. Writing 0 has no effect. Notes: [1] Normally, each CPU will clear (acknowledge) only its own local flags. This mechanism may be useful if the remote CPU is non-responsive. Reset type: SYSRSn
9	IPC9	R-0/W1S	0h	Writing 1 to this bit clears the IPC9 flag for the remote CPU. Writing 0 has no effect. Notes: [1] Normally, each CPU will clear (acknowledge) only its own local flags. This mechanism may be useful if the remote CPU is non-responsive. Reset type: SYSRSn
8	IPC8	R-0/W1S	0h	Writing 1 to this bit clears the IPC8 flag for the remote CPU. Writing 0 has no effect. Notes: [1] Normally, each CPU will clear (acknowledge) only its own local flags. This mechanism may be useful if the remote CPU is non-responsive. Reset type: SYSRSn
7	IPC7	R-0/W1S	0h	Writing 1 to this bit clears the IPC7 flag for the remote CPU. Writing 0 has no effect. Notes: [1] Normally, each CPU will clear (acknowledge) only its own local flags. This mechanism may be useful if the remote CPU is non-responsive. Reset type: SYSRSn
6	IPC6	R-0/W1S	0h	Writing 1 to this bit clears the IPC6 flag for the remote CPU. Writing 0 has no effect. Notes: [1] Normally, each CPU will clear (acknowledge) only its own local flags. This mechanism may be useful if the remote CPU is non-responsive. Reset type: SYSRSn
5	IPC5	R-0/W1S	0h	Writing 1 to this bit clears the IPC5 flag for the remote CPU. Writing 0 has no effect. Notes: [1] Normally, each CPU will clear (acknowledge) only its own local flags. This mechanism may be useful if the remote CPU is non-responsive. Reset type: SYSRSn
4	IPC4	R-0/W1S	0h	Writing 1 to this bit clears the IPC4 flag for the remote CPU. Writing 0 has no effect. Notes: [1] Normally, each CPU will clear (acknowledge) only its own local flags. This mechanism may be useful if the remote CPU is non-responsive. Reset type: SYSRSn

Table 7-9. IPCCLR Register Field Descriptions (continued)

Bit	Field	Type	Reset	Description
3	IPC3	R-0/W1S	0h	Writing 1 to this bit clears the IPC3 flag for the remote CPU. Writing 0 has no effect. Notes: [1] Normally, each CPU will clear (acknowledge) only its own local flags. This mechanism may be useful if the remote CPU is non-responsive. Reset type: SYSRSn
2	IPC2	R-0/W1S	0h	Writing 1 to this bit clears the IPC2 flag for the remote CPU. Writing 0 has no effect. Notes: [1] Normally, each CPU will clear (acknowledge) only its own local flags. This mechanism may be useful if the remote CPU is non-responsive. Reset type: SYSRSn
1	IPC1	R-0/W1S	0h	Writing 1 to this bit clears the IPC1 flag for the remote CPU. Writing 0 has no effect. Notes: [1] Normally, each CPU will clear (acknowledge) only its own local flags. This mechanism may be useful if the remote CPU is non-responsive. Reset type: SYSRSn
0	IPC0	R-0/W1S	0h	Writing 1 to this bit clears the IPC0 flag for the remote CPU. Writing 0 has no effect. Notes: [1] Normally, each CPU will clear (acknowledge) only its own local flags. This mechanism may be useful if the remote CPU is non-responsive. Reset type: SYSRSn

7.7.2.5 IPCFLG Register (Offset = 8h) [Reset = 00000000h]

IPCFLG is shown in Figure 7-6 and described in Table 7-10.

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IPC remote flag status register

Figure 7-6. IPCFLG Register

31	30	29	28	27	26	25	24
IPC31	IPC30	IPC29	IPC28	IPC27	IPC26	IPC25	IPC24
R-0h	R-0h	R-0h	R-0h	R-0h	R-0h	R-0h	R-0h
23	22	21	20	19	18	17	16
IPC23	IPC22	IPC21	IPC20	IPC19	IPC18	IPC17	IPC16
R-0h	R-0h	R-0h	R-0h	R-0h	R-0h	R-0h	R-0h
15	14	13	12	11	10	9	8
IPC15	IPC14	IPC13	IPC12	IPC11	IPC10	IPC9	IPC8
R-0h	R-0h	R-0h	R-0h	R-0h	R-0h	R-0h	R-0h
7	6	5	4	3	2	1	0
IPC7	IPC6	IPC5	IPC4	IPC3	IPC2	IPC1	IPC0
R-0h	R-0h	R-0h	R-0h	R-0h	R-0h	R-0h	R-0h

Table 7-10. IPCFLG Register Field Descriptions

Bit	Field	Type	Reset	Description
31	IPC31	R	0h	Indicates to the local CPU whether the remote IPC31 event flag is set. 0: The remote IPC31 event flag is not set 1: The remote IPC31 event flag is set Reset type: SYSRSn
30	IPC30	R	0h	Indicates to the local CPU whether the remote IPC30 event flag is set. 0: The remote IPC30 event flag is not set 1: The remote IPC30 event flag is set Reset type: SYSRSn
29	IPC29	R	0h	Indicates to the local CPU whether the remote IPC29 event flag is set. 0: The remote IPC29 event flag is not set 1: The remote IPC29 event flag is set Reset type: SYSRSn
28	IPC28	R	0h	Indicates to the local CPU whether the remote IPC28 event flag is set. 0: The remote IPC28 event flag is not set 1: The remote IPC28 event flag is set Reset type: SYSRSn
27	IPC27	R	0h	Indicates to the local CPU whether the remote IPC27 event flag is set. 0: The remote IPC27 event flag is not set 1: The remote IPC27 event flag is set Reset type: SYSRSn
26	IPC26	R	0h	Indicates to the local CPU whether the remote IPC26 event flag is set. 0: The remote IPC26 event flag is not set 1: The remote IPC26 event flag is set Reset type: SYSRSn

Table 7-10. IPCFLG Register Field Descriptions (continued)

Bit	Field	Type	Reset	Description
25	IPC25	R	0h	Indicates to the local CPU whether the remote IPC25 event flag is set. 0: The remote IPC25 event flag is not set 1: The remote IPC25 event flag is set Reset type: SYSRSn
24	IPC24	R	0h	Indicates to the local CPU whether the remote IPC24 event flag is set. 0: The remote IPC24 event flag is not set 1: The remote IPC24 event flag is set Reset type: SYSRSn
23	IPC23	R	0h	Indicates to the local CPU whether the remote IPC23 event flag is set. 0: The remote IPC23 event flag is not set 1: The remote IPC23 event flag is set Reset type: SYSRSn
22	IPC22	R	0h	Indicates to the local CPU whether the remote IPC22 event flag is set. 0: The remote IPC22 event flag is not set 1: The remote IPC22 event flag is set Reset type: SYSRSn
21	IPC21	R	0h	Indicates to the local CPU whether the remote IPC21 event flag is set. 0: The remote IPC21 event flag is not set 1: The remote IPC21 event flag is set Reset type: SYSRSn
20	IPC20	R	0h	Indicates to the local CPU whether the remote IPC20 event flag is set. 0: The remote IPC20 event flag is not set 1: The remote IPC20 event flag is set Reset type: SYSRSn
19	IPC19	R	0h	Indicates to the local CPU whether the remote IPC19 event flag is set. 0: The remote IPC19 event flag is not set 1: The remote IPC19 event flag is set Reset type: SYSRSn
18	IPC18	R	0h	Indicates to the local CPU whether the remote IPC18 event flag is set. 0: The remote IPC18 event flag is not set 1: The remote IPC18 event flag is set Reset type: SYSRSn
17	IPC17	R	0h	Indicates to the local CPU whether the remote IPC17 event flag is set. 0: The remote IPC17 event flag is not set 1: The remote IPC17 event flag is set Reset type: SYSRSn
16	IPC16	R	0h	Indicates to the local CPU whether the remote IPC16 event flag is set. 0: The remote IPC16 event flag is not set 1: The remote IPC16 event flag is set Reset type: SYSRSn
15	IPC15	R	0h	Indicates to the local CPU whether the remote IPC15 event flag is set. 0: The remote IPC15 event flag is not set 1: The remote IPC15 event flag is set Reset type: SYSRSn

Table 7-10. IPCFLG Register Field Descriptions (continued)

Bit	Field	Type	Reset	Description
14	IPC14	R	0h	Indicates to the local CPU whether the remote IPC14 event flag is set. 0: The remote IPC14 event flag is not set 1: The remote IPC14 event flag is set Reset type: SYSRSn
13	IPC13	R	0h	Indicates to the local CPU whether the remote IPC13 event flag is set. 0: The remote IPC13 event flag is not set 1: The remote IPC13 event flag is set Reset type: SYSRSn
12	IPC12	R	0h	Indicates to the local CPU whether the remote IPC12 event flag is set. 0: The remote IPC12 event flag is not set 1: The remote IPC12 event flag is set Reset type: SYSRSn
11	IPC11	R	0h	Indicates to the local CPU whether the remote IPC11 event flag is set. 0: The remote IPC11 event flag is not set 1: The remote IPC11 event flag is set Reset type: SYSRSn
10	IPC10	R	0h	Indicates to the local CPU whether the remote IPC10 event flag is set. 0: The remote IPC10 event flag is not set 1: The remote IPC10 event flag is set Reset type: SYSRSn
9	IPC9	R	0h	Indicates to the local CPU whether the remote IPC9 event flag is set. 0: The remote IPC9 event flag is not set 1: The remote IPC9 event flag is set Reset type: SYSRSn
8	IPC8	R	0h	Indicates to the local CPU whether the remote IPC8 event flag is set. 0: The remote IPC8 event flag is not set 1: The remote IPC8 event flag is set Reset type: SYSRSn
7	IPC7	R	0h	Indicates to the local CPU whether the remote IPC7 event flag is set. 0: The remote IPC7 event flag is not set 1: The remote IPC7 event flag is set Reset type: SYSRSn
6	IPC6	R	0h	Indicates to the local CPU whether the remote IPC6 event flag is set. 0: The remote IPC6 event flag is not set 1: The remote IPC6 event flag is set Reset type: SYSRSn
5	IPC5	R	0h	Indicates to the local CPU whether the remote IPC5 event flag is set. 0: The remote IPC5 event flag is not set 1: The remote IPC5 event flag is set Reset type: SYSRSn
4	IPC4	R	0h	Indicates to the local CPU whether the remote IPC4 event flag is set. 0: The remote IPC4 event flag is not set 1: The remote IPC4 event flag is set Reset type: SYSRSn
3	IPC3	R	0h	Indicates to the local CPU whether the remote IPC3 event flag is set. 0: The remote IPC3 event flag is not set 1: The remote IPC3 event flag is set Notes: [1] IPC event flags 0-3 will trigger interrupts in the receiving CPU via the ePIE. Reset type: SYSRSn

Table 7-10. IPCFLG Register Field Descriptions (continued)

Bit	Field	Type	Reset	Description
2	IPC2	R	0h	Indicates to the local CPU whether the remote IPC2 event flag is set. 0: The remote IPC2 event flag is not set 1: The remote IPC2 event flag is set Notes: [1] IPC event flags 0-3 will trigger interrupts in the receiving CPU via the ePIE. Reset type: SYSRSn
1	IPC1	R	0h	Indicates to the local CPU whether the remote IPC1 event flag is set. 0: The remote IPC1 event flag is not set 1: The remote IPC1 event flag is set Notes: [1] IPC event flags 0-3 will trigger interrupts in the receiving CPU via the ePIE. Reset type: SYSRSn
0	IPC0	R	0h	Indicates to the local CPU whether the remote IPC0 event flag is set. 0: The remote IPC0 event flag is not set 1: The remote IPC0 event flag is set Notes: [1] IPC event flags 0-3 will trigger interrupts in the receiving CPU via the ePIE. Reset type: SYSRSn

7.7.2.6 IPCCOUNTERL Register (Offset = Ch) [Reset = 00000000h]

IPCCOUNTERL is shown in Figure 7-7 and described in Table 7-11.

Return to the Summary Table.

IPC Counter Low Register

Figure 7-7. IPCCOUNTERL Register

31–0
COUNT
R-0h

Table 7-11. IPCCOUNTERL Register Field Descriptions

Bit	Field	Type	Reset	Description
31-0	COUNT	R	0h	This is the lower 32-bits of free running 64 bit timestamp counter clocked by the PLLSYSCLK. Reset type: CPU1.SYSRSn

7.7.2.7 IPCCOUNTERH Register (Offset = Eh) [Reset = 00000000h]

IPCCOUNTERH is shown in Figure 7-8 and described in Table 7-12.

Return to the Summary Table.

IPC Counter High Register

Figure 7-8. IPCCOUNTERH Register

31–0
COUNT
R-0h

Table 7-12. IPCCOUNTERH Register Field Descriptions

Bit	Field	Type	Reset	Description
31-0	COUNT	R	0h	This is the upper 32-bits of free running 64 bit timestamp counter clocked by the PLLSYSCLK. Reset type: CPU1.SYSRSn

7.7.2.8 IPCSENDCOM Register (Offset = 10h) [Reset = 00000000h]

IPCSENDCOM is shown in Figure 7-9 and described in Table 7-13.

Return to the Summary Table.

Local to Remote IPC Command Register

Figure 7-9. IPCSENDCOM Register

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
COMMAND																															
R/W-0h																															

Table 7-13. IPCSENDCOM Register Field Descriptions

Bit	Field	Type	Reset	Description
31-0	COMMAND	R/W	0h	This is a general purpose register used to send software-defined commands to the remote CPU. It can only be written by the local CPU. Notes [1] The local CPU's IPCSENDCOM is the same physical register as the remote CPU's IPCRECVCOM, and is located at the same address in both CPUs. Reset type: SYSRSn

7.7.2.9 IPCSENDADDR Register (Offset = 12h) [Reset = 00000000h]

IPCSENDADDR is shown in Figure 7-10 and described in Table 7-14.

Return to the Summary Table.

Local to Remote IPC Address Register

Figure 7-10. IPCSENDADDR Register

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ADDRESS																															
R/W-0h																															

Table 7-14. IPCSENDADDR Register Field Descriptions

Bit	Field	Type	Reset	Description
31-0	ADDRESS	R/W	0h	This is a general purpose register used to send software-defined addresses to the remote CPU. It can only be written by the local CPU. Notes [1] The local CPU's IPCSENDADDR is the same physical register as the remote CPU's IPCRECVDATA, and is located at the same address in both CPUs. Reset type: SYSRSn

7.7.2.10 IPCSENDDATA Register (Offset = 14h) [Reset = 00000000h]

IPCSENDDATA is shown in Figure 7-11 and described in Table 7-15.

Return to the Summary Table.

Local to Remote IPC Data Register

Figure 7-11. IPCSENDDATA Register

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
WDATA																															
R/W-0h																															

Table 7-15. IPCSENDDATA Register Field Descriptions

Bit	Field	Type	Reset	Description
31-0	WDATA	R/W	0h	This is a general purpose register used to send software-defined data to the remote CPU. It can only be written by the local CPU. Notes [1] The local CPU's IPCSENDDATA is the same physical register as the remote CPU's IPCRECVDATA, and is located at the same address in both CPUs. Reset type: SYSRSn

7.7.2.11 IPCREMOTEREPLY Register (Offset = 16h) [Reset = 00000000h]

IPCREMOTEREPLY is shown in Figure 7-12 and described in Table 7-16.

Return to the Summary Table.

Remote to Local IPC Reply Data Register

Figure 7-12. IPCREMOTEREPLY Register

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RDATA																															
R-0h																															

Table 7-16. IPCREMOTEREPLY Register Field Descriptions

Bit	Field	Type	Reset	Description
31-0	RDATA	R	0h	This is a general purpose register used to receive software-defined data from the remote CPU's response to a command. It can only be written by the remote CPU. Notes [1] The local CPU's IPCREMOTEREPLY is the same physical register as the remote CPU's IPCLOCALREPLY, and is located at the same address in both CPUs. [2] This register is reset by a SYRSn of the remote CPU Reset type: CPUx.SYSRSn

7.7.2.12 IPCRECVCOM Register (Offset = 18h) [Reset = 00000000h]

IPCRECVCOM is shown in Figure 7-13 and described in Table 7-17.

Return to the Summary Table.

Remote to Local IPC Command Register

Figure 7-13. IPCRECVCOM Register

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
COMMAND																															
R-0h																															

Table 7-17. IPCRECVCOM Register Field Descriptions

Bit	Field	Type	Reset	Description
31-0	COMMAND	R	0h	This is a general purpose register used to receive software-defined commands from the remote CPU. It can only be written by the remote CPU. Notes [1] The local CPU's IPCRECVCOM is the same physical register as the remote CPU's IPCSENDCOM, and is located at the same address in both CPUs. [2] This register is reset by a SYRSn of the remote CPU Reset type: CPUx.SYSRSn

7.7.2.13 IPCRECVADDR Register (Offset = 1Ah) [Reset = 00000000h]

IPCRECVADDR is shown in Figure 7-14 and described in Table 7-18.

Return to the Summary Table.

Remote to Local IPC Address Register

Figure 7-14. IPCRECVADDR Register

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ADDRESS																															
R-0h																															

Table 7-18. IPCRECVADDR Register Field Descriptions

Bit	Field	Type	Reset	Description
31-0	ADDRESS	R	0h	This is a general purpose register used to receive software-defined addresses from the remote CPU. It can only be written by the remote CPU. Notes [1] The local CPU's IPCRECVADDR is the same physical register as the remote CPU's IPCSENDADDR, and is located at the same address in both CPUs. [2] This register is reset by a SYRSn of the remote CPU Reset type: CPUx.SYSRSn

7.7.2.14 IPCRECVDATA Register (Offset = 1Ch) [Reset = 00000000h]

IPCRECVDATA is shown in Figure 7-15 and described in Table 7-19.

Return to the Summary Table.

Remote to Local IPC Data Register

Figure 7-15. IPCRECVDATA Register

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
WDATA																															
R-0h																															

Table 7-19. IPCRECVDATA Register Field Descriptions

Bit	Field	Type	Reset	Description
31-0	WDATA	R	0h	This is a general purpose register used to receive software-defined data from the remote CPU. It can only be written by the remote CPU. Notes [1] The local CPU's IPCRECVDATA is the same physical register as the remote CPU's IPCSENDDATA, and is located at the same address in both CPUs. [2] This register is reset by a SYRSn of the remote CPU Reset type: CPUx.SYSRSn

7.7.2.15 IPCLOCALREPLY Register (Offset = 1Eh) [Reset = 00000000h]

IPCLOCALREPLY is shown in Figure 7-16 and described in Table 7-20.

Return to the Summary Table.

Local to Remote IPC Reply Data Register

Figure 7-16. IPCLOCALREPLY Register

31–0
RDATA
R/W-0h

Table 7-20. IPCLOCALREPLY Register Field Descriptions

Bit	Field	Type	Reset	Description
31-0	RDATA	R/W	0h	This is a general purpose register used to send software-defined data to the remote CPU in response to a command. It can only be written by the local CPU. Notes [1] The local CPU's IPCLOCALREPLY is the same physical register as the remote CPU's IPCREMOTEREPLY, and is located at the same address in both CPUs. Reset type: SYSRSn

7.7.2.16 IPCBOOTSTS Register (Offset = 20h) [Reset = 00000000h]

IPCBOOTSTS is shown in Figure 7-17 and described in Table 7-21.

Return to the Summary Table.

CPU2 to CPU1 IPC Boot Status Register

Figure 7-17. IPCBOOTSTS Register

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
BOOTSTS																															
R/W-0h																															

Table 7-21. IPCBOOTSTS Register Field Descriptions

Bit	Field	Type	Reset	Description
31-0	BOOTSTS	R/W	0h	This register is used by CPU2 to pass the boot Status to CPU1. The data format is software-defined. It can only be written by CPU2. Reset type: CPU2.SYSRSn

7.7.2.17 IPCBOOTMODE Register (Offset = 22h) [Reset = 00000000h]

IPCBOOTMODE is shown in Figure 7-18 and described in Table 7-22.

Return to the Summary Table.

CPU1 to CPU2 IPC Boot Mode Register

Figure 7-18. IPCBOOTMODE Register

31-0
BOOTMODE
R/W-0h

Table 7-22. IPCBOOTMODE Register Field Descriptions

Bit	Field	Type	Reset	Description
31-0	BOOTMODE	R/W	0h	This register is used by CPU1 to pass a boot mode information to CPU2. The data format is software-defined. It can only be written by CPU1. Reset type: CPU1.SYSRSn

7.7.3 IPC_REGS_CPU2 Registers

Table 7-23 lists the memory-mapped registers for the IPC_REGS_CPU2 registers. All register offset addresses not listed in Table 7-23 should be considered as reserved locations and the register contents should not be modified.

Table 7-23. IPC_REGS_CPU2 Registers

Offset	Acronym	Register Name	Write Protection	Section
0h	IPCACK	IPC incoming flag clear (acknowledge) register		Go
2h	IPCSTS	IPC incoming flag status register		Go
4h	IPCSET	IPC remote flag set register		Go
6h	IPCCLR	IPC remote flag clear register		Go
8h	IPCFLG	IPC remote flag status register		Go
Ch	IPCCOUNTERL	IPC Counter Low Register		Go
Eh	IPCCOUNTERH	IPC Counter High Register		Go
10h	IPCRECVCOM	Remote to Local IPC Command Register		Go
12h	IPCRECVADDR	Remote to Local IPC Address Register		Go
14h	IPCRECVDATA	Remote to Local IPC Data Register		Go
16h	IPCLOCALREPLY	Local to Remote IPC Reply Data Register		Go
18h	IPCSENDCOM	Local to Remote IPC Command Register		Go
1Ah	IPCSENDADDR	Local to Remote IPC Address Register		Go
1Ch	IPCSENDDATA	Local to Remote IPC Data Register		Go
1Eh	IPCREMOTEREPLY	Remote to Local IPC Reply Data Register		Go
20h	IPCBOOTSTS	CPU2 to CPU1 IPC Boot Status Register		Go
22h	IPCBOOTMODE	CPU1 to CPU2 IPC Boot Mode Register		Go

Complex bit access types are encoded to fit into small table cells. Table 7-24 shows the codes that are used for access types in this section.

Table 7-24. IPC_REGS_CPU2 Access Type Codes

Access Type	Code	Description
Read Type		
R	R	Read
R-0	R -0	Read Returns 0s
Write Type		
W	W	Write
W1S	W 1S	Write 1 to set
Reset or Default Value		
-n		Value after reset or the default value
Register Array Variables		
i,j,k,l,m,n		When these variables are used in a register name, an offset, or an address, they refer to the value of a register array where the register is part of a group of repeating registers. The register groups form a hierarchical structure and the array is represented with a formula.
y		When this variable is used in a register name, an offset, or an address it refers to the value of a register array.

7.7.3.1 IPCACK Register (Offset = 0h) [Reset = 00000000h]

IPCACK is shown in Figure 7-19 and described in Table 7-25.

Return to the Summary Table.

IPC incoming flag clear (acknowledge) register

Figure 7-19. IPCACK Register

31	30	29	28	27	26	25	24
IPC31	IPC30	IPC29	IPC28	IPC27	IPC26	IPC25	IPC24
R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h
23	22	21	20	19	18	17	16
IPC23	IPC22	IPC21	IPC20	IPC19	IPC18	IPC17	IPC16
R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h
15	14	13	12	11	10	9	8
IPC15	IPC14	IPC13	IPC12	IPC11	IPC10	IPC9	IPC8
R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h
7	6	5	4	3	2	1	0
IPC7	IPC6	IPC5	IPC4	IPC3	IPC2	IPC1	IPC0
R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h

Table 7-25. IPCACK Register Field Descriptions

Bit	Field	Type	Reset	Description
31	IPC31	R-0/W1S	0h	Writing 1 to this bit clears the IPC31 event flag which was set by the remote CPU. Writing 0 to this bit has no effect. Reset type: SYSRSn
30	IPC30	R-0/W1S	0h	Writing 1 to this bit clears the IPC30 event flag which was set by the remote CPU. Writing 0 to this bit has no effect. Reset type: SYSRSn
29	IPC29	R-0/W1S	0h	Writing 1 to this bit clears the IPC29 event flag which was set by the remote CPU. Writing 0 to this bit has no effect. Reset type: SYSRSn
28	IPC28	R-0/W1S	0h	Writing 1 to this bit clears the IPC28 event flag which was set by the remote CPU. Writing 0 to this bit has no effect. Reset type: SYSRSn
27	IPC27	R-0/W1S	0h	Writing 1 to this bit clears the IPC27 event flag which was set by the remote CPU. Writing 0 to this bit has no effect. Reset type: SYSRSn
26	IPC26	R-0/W1S	0h	Writing 1 to this bit clears the IPC26 event flag which was set by the remote CPU. Writing 0 to this bit has no effect. Reset type: SYSRSn
25	IPC25	R-0/W1S	0h	Writing 1 to this bit clears the IPC25 event flag which was set by the remote CPU. Writing 0 to this bit has no effect. Reset type: SYSRSn

Table 7-25. IPCACK Register Field Descriptions (continued)

Bit	Field	Type	Reset	Description
24	IPC24	R-0/W1S	0h	Writing 1 to this bit clears the IPC24 event flag which was set by the remote CPU. Writing 0 to this bit has no effect. Reset type: SYSRSn
23	IPC23	R-0/W1S	0h	Writing 1 to this bit clears the IPC23 event flag which was set by the remote CPU. Writing 0 to this bit has no effect. Reset type: SYSRSn
22	IPC22	R-0/W1S	0h	Writing 1 to this bit clears the IPC22 event flag which was set by the remote CPU. Writing 0 to this bit has no effect. Reset type: SYSRSn
21	IPC21	R-0/W1S	0h	Writing 1 to this bit clears the IPC21 event flag which was set by the remote CPU. Writing 0 to this bit has no effect. Reset type: SYSRSn
20	IPC20	R-0/W1S	0h	Writing 1 to this bit clears the IPC20 event flag which was set by the remote CPU. Writing 0 to this bit has no effect. Reset type: SYSRSn
19	IPC19	R-0/W1S	0h	Writing 1 to this bit clears the IPC19 event flag which was set by the remote CPU. Writing 0 to this bit has no effect. Reset type: SYSRSn
18	IPC18	R-0/W1S	0h	Writing 1 to this bit clears the IPC18 event flag which was set by the remote CPU. Writing 0 to this bit has no effect. Reset type: SYSRSn
17	IPC17	R-0/W1S	0h	Writing 1 to this bit clears the IPC17 event flag which was set by the remote CPU. Writing 0 to this bit has no effect. Reset type: SYSRSn
16	IPC16	R-0/W1S	0h	Writing 1 to this bit clears the IPC16 event flag which was set by the remote CPU. Writing 0 to this bit has no effect. Reset type: SYSRSn
15	IPC15	R-0/W1S	0h	Writing 1 to this bit clears the IPC15 event flag which was set by the remote CPU. Writing 0 to this bit has no effect. Reset type: SYSRSn
14	IPC14	R-0/W1S	0h	Writing 1 to this bit clears the IPC14 event flag which was set by the remote CPU. Writing 0 to this bit has no effect. Reset type: SYSRSn
13	IPC13	R-0/W1S	0h	Writing 1 to this bit clears the IPC13 event flag which was set by the remote CPU. Writing 0 to this bit has no effect. Reset type: SYSRSn
12	IPC12	R-0/W1S	0h	Writing 1 to this bit clears the IPC12 event flag which was set by the remote CPU. Writing 0 to this bit has no effect. Reset type: SYSRSn
11	IPC11	R-0/W1S	0h	Writing 1 to this bit clears the IPC11 event flag which was set by the remote CPU. Writing 0 to this bit has no effect. Reset type: SYSRSn

Table 7-25. IPCACK Register Field Descriptions (continued)

Bit	Field	Type	Reset	Description
10	IPC10	R-0/W1S	0h	Writing 1 to this bit clears the IPC10 event flag which was set by the remote CPU. Writing 0 to this bit has no effect. Reset type: SYSRSn
9	IPC9	R-0/W1S	0h	Writing 1 to this bit clears the IPC9 event flag which was set by the remote CPU. Writing 0 to this bit has no effect. Reset type: SYSRSn
8	IPC8	R-0/W1S	0h	Writing 1 to this bit clears the IPC8 event flag which was set by the remote CPU. Writing 0 to this bit has no effect. Reset type: SYSRSn
7	IPC7	R-0/W1S	0h	Writing 1 to this bit clears the IPC7 event flag which was set by the remote CPU. Writing 0 to this bit has no effect. Reset type: SYSRSn
6	IPC6	R-0/W1S	0h	Writing 1 to this bit clears the IPC6 event flag which was set by the remote CPU. Writing 0 to this bit has no effect. Reset type: SYSRSn
5	IPC5	R-0/W1S	0h	Writing 1 to this bit clears the IPC5 event flag which was set by the remote CPU. Writing 0 to this bit has no effect. Reset type: SYSRSn
4	IPC4	R-0/W1S	0h	Writing 1 to this bit clears the IPC4 event flag which was set by the remote CPU. Writing 0 to this bit has no effect. Reset type: SYSRSn
3	IPC3	R-0/W1S	0h	Writing 1 to this bit clears the IPC3 event flag which was set by the remote CPU. Writing 0 to this bit has no effect. Reset type: SYSRSn
2	IPC2	R-0/W1S	0h	Writing 1 to this bit clears the IPC2 event flag which was set by the remote CPU. Writing 0 to this bit has no effect. Reset type: SYSRSn
1	IPC1	R-0/W1S	0h	Writing 1 to this bit clears the IPC1 event flag which was set by the remote CPU. Writing 0 to this bit has no effect. Reset type: SYSRSn
0	IPC0	R-0/W1S	0h	Writing 1 to this bit clears the IPC0 event flag which was set by the remote CPU. Writing 0 to this bit has no effect. Reset type: SYSRSn

7.7.3.2 IPCSTS Register (Offset = 2h) [Reset = 00000000h]

IPCSTS is shown in Figure 7-20 and described in Table 7-26.

Return to the Summary Table.

IPC incoming flag status register

Figure 7-20. IPCSTS Register

31	30	29	28	27	26	25	24
IPC31	IPC30	IPC29	IPC28	IPC27	IPC26	IPC25	IPC24
R-0h	R-0h	R-0h	R-0h	R-0h	R-0h	R-0h	R-0h
23	22	21	20	19	18	17	16
IPC23	IPC22	IPC21	IPC20	IPC19	IPC18	IPC17	IPC16
R-0h	R-0h	R-0h	R-0h	R-0h	R-0h	R-0h	R-0h
15	14	13	12	11	10	9	8
IPC15	IPC14	IPC13	IPC12	IPC11	IPC10	IPC9	IPC8
R-0h	R-0h	R-0h	R-0h	R-0h	R-0h	R-0h	R-0h
7	6	5	4	3	2	1	0
IPC7	IPC6	IPC5	IPC4	IPC3	IPC2	IPC1	IPC0
R-0h	R-0h	R-0h	R-0h	R-0h	R-0h	R-0h	R-0h

Table 7-26. IPCSTS Register Field Descriptions

Bit	Field	Type	Reset	Description
31	IPC31	R	0h	Indicates to the local CPU if the IPC31 event flag was set by the remote CPU. 0: No IPC31 event was set by the remote CPU 1: An IPC31 event was set by the remote CPU Reset type: SYSRSn
30	IPC30	R	0h	Indicates to the local CPU if the IPC30 event flag was set by the remote CPU. 0: No IPC30 event was set by the remote CPU 1: An IPC30 event was set by the remote CPU Reset type: SYSRSn
29	IPC29	R	0h	Indicates to the local CPU if the IPC29 event flag was set by the remote CPU. 0: No IPC29 event was set by the remote CPU 1: An IPC29 event was set by the remote CPU Reset type: SYSRSn
28	IPC28	R	0h	Indicates to the local CPU if the IPC28 event flag was set by the remote CPU. 0: No IPC28 event was set by the remote CPU 1: An IPC28 event was set by the remote CPU Reset type: SYSRSn
27	IPC27	R	0h	Indicates to the local CPU if the IPC27 event flag was set by the remote CPU. 0: No IPC27 event was set by the remote CPU 1: An IPC27 event was set by the remote CPU Reset type: SYSRSn
26	IPC26	R	0h	Indicates to the local CPU if the IPC26 event flag was set by the remote CPU. 0: No IPC26 event was set by the remote CPU 1: An IPC26 event was set by the remote CPU Reset type: SYSRSn

Table 7-26. IPCSTS Register Field Descriptions (continued)

Bit	Field	Type	Reset	Description
25	IPC25	R	0h	Indicates to the local CPU if the IPC25 event flag was set by the remote CPU. 0: No IPC25 event was set by the remote CPU 1: An IPC25 event was set by the remote CPU Reset type: SYSRSn
24	IPC24	R	0h	Indicates to the local CPU if the IPC24 event flag was set by the remote CPU. 0: No IPC24 event was set by the remote CPU 1: An IPC24 event was set by the remote CPU Reset type: SYSRSn
23	IPC23	R	0h	Indicates to the local CPU if the IPC23 event flag was set by the remote CPU. 0: No IPC23 event was set by the remote CPU 1: An IPC23 event was set by the remote CPU Reset type: SYSRSn
22	IPC22	R	0h	Indicates to the local CPU if the IPC22 event flag was set by the remote CPU. 0: No IPC22 event was set by the remote CPU 1: An IPC22 event was set by the remote CPU Reset type: SYSRSn
21	IPC21	R	0h	Indicates to the local CPU if the IPC21 event flag was set by the remote CPU. 0: No IPC21 event was set by the remote CPU 1: An IPC21 event was set by the remote CPU Reset type: SYSRSn
20	IPC20	R	0h	Indicates to the local CPU if the IPC20 event flag was set by the remote CPU. 0: No IPC20 event was set by the remote CPU 1: An IPC20 event was set by the remote CPU Reset type: SYSRSn
19	IPC19	R	0h	Indicates to the local CPU if the IPC19 event flag was set by the remote CPU. 0: No IPC19 event was set by the remote CPU 1: An IPC19 event was set by the remote CPU Reset type: SYSRSn
18	IPC18	R	0h	Indicates to the local CPU if the IPC18 event flag was set by the remote CPU. 0: No IPC18 event was set by the remote CPU 1: An IPC18 event was set by the remote CPU Reset type: SYSRSn
17	IPC17	R	0h	Indicates to the local CPU if the IPC17 event flag was set by the remote CPU. 0: No IPC17 event was set by the remote CPU 1: An IPC17 event was set by the remote CPU Reset type: SYSRSn
16	IPC16	R	0h	Indicates to the local CPU if the IPC16 event flag was set by the remote CPU. 0: No IPC16 event was set by the remote CPU 1: An IPC16 event was set by the remote CPU Reset type: SYSRSn
15	IPC15	R	0h	Indicates to the local CPU if the IPC15 event flag was set by the remote CPU. 0: No IPC15 event was set by the remote CPU 1: An IPC15 event was set by the remote CPU Reset type: SYSRSn

Table 7-26. IPCSTS Register Field Descriptions (continued)

Bit	Field	Type	Reset	Description
14	IPC14	R	0h	Indicates to the local CPU if the IPC14 event flag was set by the remote CPU. 0: No IPC14 event was set by the remote CPU 1: An IPC14 event was set by the remote CPU Reset type: SYSRSn
13	IPC13	R	0h	Indicates to the local CPU if the IPC13 event flag was set by the remote CPU. 0: No IPC13 event was set by the remote CPU 1: An IPC13 event was set by the remote CPU Reset type: SYSRSn
12	IPC12	R	0h	Indicates to the local CPU if the IPC12 event flag was set by the remote CPU. 0: No IPC12 event was set by the remote CPU 1: An IPC12 event was set by the remote CPU Reset type: SYSRSn
11	IPC11	R	0h	Indicates to the local CPU if the IPC11 event flag was set by the remote CPU. 0: No IPC11 event was set by the remote CPU 1: An IPC11 event was set by the remote CPU Reset type: SYSRSn
10	IPC10	R	0h	Indicates to the local CPU if the IPC10 event flag was set by the remote CPU. 0: No IPC10 event was set by the remote CPU 1: An IPC10 event was set by the remote CPU Reset type: SYSRSn
9	IPC9	R	0h	Indicates to the local CPU if the IPC9 event flag was set by the remote CPU. 0: No IPC9 event was set by the remote CPU 1: An IPC9 event was set by the remote CPU Reset type: SYSRSn
8	IPC8	R	0h	Indicates to the local CPU if the IPC8 event flag was set by the remote CPU. 0: No IPC8 event was set by the remote CPU 1: An IPC8 event was set by the remote CPU Reset type: SYSRSn
7	IPC7	R	0h	Indicates to the local CPU if the IPC7 event flag was set by the remote CPU. 0: No IPC7 event was set by the remote CPU 1: An IPC7 event was set by the remote CPU Reset type: SYSRSn
6	IPC6	R	0h	Indicates to the local CPU if the IPC6 event flag was set by the remote CPU. 0: No IPC6 event was set by the remote CPU 1: An IPC6 event was set by the remote CPU Reset type: SYSRSn
5	IPC5	R	0h	Indicates to the local CPU if the IPC5 event flag was set by the remote CPU. 0: No IPC5 event was set by the remote CPU 1: An IPC5 event was set by the remote CPU Reset type: SYSRSn
4	IPC4	R	0h	Indicates to the local CPU if the IPC4 event flag was set by the remote CPU. 0: No IPC4 event was set by the remote CPU 1: An IPC4 event was set by the remote CPU Reset type: SYSRSn

Table 7-26. IPCSTS Register Field Descriptions (continued)

Bit	Field	Type	Reset	Description
3	IPC3	R	0h	Indicates to the local CPU if the IPC3 event flag was set by the remote CPU. 0: No IPC3 event was set by the remote CPU 1: An IPC3 event was set by the remote CPU Notes [1] IPC event flags 0-3 will trigger interrupts in the receiving CPU via the ePIE. Reset type: SYSRSn
2	IPC2	R	0h	Indicates to the local CPU if the IPC2 event flag was set by the remote CPU. 0: No IPC2 event was set by the remote CPU 1: An IPC2 event was set by the remote CPU Notes [1] IPC event flags 0-3 will trigger interrupts in the receiving CPU via the ePIE. Reset type: SYSRSn
1	IPC1	R	0h	Indicates to the local CPU if the IPC1 event flag was set by the remote CPU. 0: No IPC1 event was set by the remote CPU 1: An IPC1 event was set by the remote CPU Notes [1] IPC event flags 0-3 will trigger interrupts in the receiving CPU via the ePIE. Reset type: SYSRSn
0	IPC0	R	0h	Indicates to the local CPU if the IPC0 event flag was set by the remote CPU. 0: No IPC0 event was set by the remote CPU 1: An IPC0 event was set by the remote CPU Notes [1] IPC event flags 0-3 will trigger interrupts in the receiving CPU via the ePIE. Reset type: SYSRSn

7.7.3.3 IPCSET Register (Offset = 4h) [Reset = 00000000h]

IPCSET is shown in Figure 7-21 and described in Table 7-27.

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IPC remote flag set register

Figure 7-21. IPCSET Register

31	30	29	28	27	26	25	24
IPC31	IPC30	IPC29	IPC28	IPC27	IPC26	IPC25	IPC24
R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h
23	22	21	20	19	18	17	16
IPC23	IPC22	IPC21	IPC20	IPC19	IPC18	IPC17	IPC16
R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h
15	14	13	12	11	10	9	8
IPC15	IPC14	IPC13	IPC12	IPC11	IPC10	IPC9	IPC8
R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h
7	6	5	4	3	2	1	0
IPC7	IPC6	IPC5	IPC4	IPC3	IPC2	IPC1	IPC0
R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h

Table 7-27. IPCSET Register Field Descriptions

Bit	Field	Type	Reset	Description
31	IPC31	R-0/W1S	0h	Writing 1 to this bit sets the IPC31 event flag for the remote CPU. Writing 0 has no effect. Reset type: SYSRSn
30	IPC30	R-0/W1S	0h	Writing 1 to this bit sets the IPC30 event flag for the remote CPU. Writing 0 has no effect. Reset type: SYSRSn
29	IPC29	R-0/W1S	0h	Writing 1 to this bit sets the IPC29 event flag for the remote CPU. Writing 0 has no effect. Reset type: SYSRSn
28	IPC28	R-0/W1S	0h	Writing 1 to this bit sets the IPC28 event flag for the remote CPU. Writing 0 has no effect. Reset type: SYSRSn
27	IPC27	R-0/W1S	0h	Writing 1 to this bit sets the IPC27 event flag for the remote CPU. Writing 0 has no effect. Reset type: SYSRSn
26	IPC26	R-0/W1S	0h	Writing 1 to this bit sets the IPC26 event flag for the remote CPU. Writing 0 has no effect. Reset type: SYSRSn
25	IPC25	R-0/W1S	0h	Writing 1 to this bit sets the IPC25 event flag for the remote CPU. Writing 0 has no effect. Reset type: SYSRSn
24	IPC24	R-0/W1S	0h	Writing 1 to this bit sets the IPC24 event flag for the remote CPU. Writing 0 has no effect. Reset type: SYSRSn
23	IPC23	R-0/W1S	0h	Writing 1 to this bit sets the IPC23 event flag for the remote CPU. Writing 0 has no effect. Reset type: SYSRSn
22	IPC22	R-0/W1S	0h	Writing 1 to this bit sets the IPC22 event flag for the remote CPU. Writing 0 has no effect. Reset type: SYSRSn

Table 7-27. IPCSET Register Field Descriptions (continued)

Bit	Field	Type	Reset	Description
21	IPC21	R-0/W1S	0h	Writing 1 to this bit sets the IPC21 event flag for the remote CPU. Writing 0 has no effect. Reset type: SYSRSn
20	IPC20	R-0/W1S	0h	Writing 1 to this bit sets the IPC20 event flag for the remote CPU. Writing 0 has no effect. Reset type: SYSRSn
19	IPC19	R-0/W1S	0h	Writing 1 to this bit sets the IPC19 event flag for the remote CPU. Writing 0 has no effect. Reset type: SYSRSn
18	IPC18	R-0/W1S	0h	Writing 1 to this bit sets the IPC18 event flag for the remote CPU. Writing 0 has no effect. Reset type: SYSRSn
17	IPC17	R-0/W1S	0h	Writing 1 to this bit sets the IPC17 event flag for the remote CPU. Writing 0 has no effect. Reset type: SYSRSn
16	IPC16	R-0/W1S	0h	Writing 1 to this bit sets the IPC16 event flag for the remote CPU. Writing 0 has no effect. Reset type: SYSRSn
15	IPC15	R-0/W1S	0h	Writing 1 to this bit sets the IPC15 event flag for the remote CPU. Writing 0 has no effect. Reset type: SYSRSn
14	IPC14	R-0/W1S	0h	Writing 1 to this bit sets the IPC14 event flag for the remote CPU. Writing 0 has no effect. Reset type: SYSRSn
13	IPC13	R-0/W1S	0h	Writing 1 to this bit sets the IPC13 event flag for the remote CPU. Writing 0 has no effect. Reset type: SYSRSn
12	IPC12	R-0/W1S	0h	Writing 1 to this bit sets the IPC12 event flag for the remote CPU. Writing 0 has no effect. Reset type: SYSRSn
11	IPC11	R-0/W1S	0h	Writing 1 to this bit sets the IPC11 event flag for the remote CPU. Writing 0 has no effect. Reset type: SYSRSn
10	IPC10	R-0/W1S	0h	Writing 1 to this bit sets the IPC10 event flag for the remote CPU. Writing 0 has no effect. Reset type: SYSRSn
9	IPC9	R-0/W1S	0h	Writing 1 to this bit sets the IPC9 event flag for the remote CPU. Writing 0 has no effect. Reset type: SYSRSn
8	IPC8	R-0/W1S	0h	Writing 1 to this bit sets the IPC8 event flag for the remote CPU. Writing 0 has no effect. Reset type: SYSRSn
7	IPC7	R-0/W1S	0h	Writing 1 to this bit sets the IPC7 event flag for the remote CPU. Writing 0 has no effect. Reset type: SYSRSn
6	IPC6	R-0/W1S	0h	Writing 1 to this bit sets the IPC6 event flag for the remote CPU. Writing 0 has no effect. Reset type: SYSRSn
5	IPC5	R-0/W1S	0h	Writing 1 to this bit sets the IPC5 event flag for the remote CPU. Writing 0 has no effect. Reset type: SYSRSn

Table 7-27. IPCSET Register Field Descriptions (continued)

Bit	Field	Type	Reset	Description
4	IPC4	R-0/W1S	0h	Writing 1 to this bit sets the IPC4 event flag for the remote CPU. Writing 0 has no effect. Reset type: SYSRSn
3	IPC3	R-0/W1S	0h	Writing 1 to this bit sets the IPC3 event flag for the remote CPU. Writing 0 has no effect. Notes: [1] IPC event flags 0-3 will trigger interrupts in the receiving CPU via the ePIE. Reset type: SYSRSn
2	IPC2	R-0/W1S	0h	Writing 1 to this bit sets the IPC2 event flag for the remote CPU. Writing 0 has no effect. Notes: [1] IPC event flags 0-3 will trigger interrupts in the receiving CPU via the ePIE. Reset type: SYSRSn
1	IPC1	R-0/W1S	0h	Writing 1 to this bit sets the IPC1 event flag for the remote CPU. Writing 0 has no effect. Notes: [1] IPC event flags 0-3 will trigger interrupts in the receiving CPU via the ePIE. Reset type: SYSRSn
0	IPC0	R-0/W1S	0h	Writing 1 to this bit sets the IPC0 event flag for the remote CPU. Writing 0 has no effect. Notes: [1] IPC event flags 0-3 will trigger interrupts in the receiving CPU via the ePIE. Reset type: SYSRSn

7.7.3.4 IPCCLR Register (Offset = 6h) [Reset = 00000000h]

IPCCLR is shown in Figure 7-22 and described in Table 7-28.

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IPC remote flag clear register

Figure 7-22. IPCCLR Register

31	30	29	28	27	26	25	24
IPC31	IPC30	IPC29	IPC28	IPC27	IPC26	IPC25	IPC24
R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h
23	22	21	20	19	18	17	16
IPC23	IPC22	IPC21	IPC20	IPC19	IPC18	IPC17	IPC16
R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h
15	14	13	12	11	10	9	8
IPC15	IPC14	IPC13	IPC12	IPC11	IPC10	IPC9	IPC8
R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h
7	6	5	4	3	2	1	0
IPC7	IPC6	IPC5	IPC4	IPC3	IPC2	IPC1	IPC0
R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h	R-0/W1S-0h

Table 7-28. IPCCLR Register Field Descriptions

Bit	Field	Type	Reset	Description
31	IPC31	R-0/W1S	0h	Writing 1 to this bit clears the IPC31 flag for the remote CPU. Writing 0 has no effect. Notes: [1] Normally, each CPU will clear (acknowledge) only its own local flags. This mechanism may be useful if the remote CPU is non-responsive. Reset type: SYSRSn
30	IPC30	R-0/W1S	0h	Writing 1 to this bit clears the IPC30 flag for the remote CPU. Writing 0 has no effect. Notes: [1] Normally, each CPU will clear (acknowledge) only its own local flags. This mechanism may be useful if the remote CPU is non-responsive. Reset type: SYSRSn
29	IPC29	R-0/W1S	0h	Writing 1 to this bit clears the IPC29 flag for the remote CPU. Writing 0 has no effect. Notes: [1] Normally, each CPU will clear (acknowledge) only its own local flags. This mechanism may be useful if the remote CPU is non-responsive. Reset type: SYSRSn
28	IPC28	R-0/W1S	0h	Writing 1 to this bit clears the IPC28 flag for the remote CPU. Writing 0 has no effect. Notes: [1] Normally, each CPU will clear (acknowledge) only its own local flags. This mechanism may be useful if the remote CPU is non-responsive. Reset type: SYSRSn

Table 7-28. IPCCLR Register Field Descriptions (continued)

Bit	Field	Type	Reset	Description
27	IPC27	R-0/W1S	0h	Writing 1 to this bit clears the IPC27 flag for the remote CPU. Writing 0 has no effect. Notes: [1] Normally, each CPU will clear (acknowledge) only its own local flags. This mechanism may be useful if the remote CPU is non-responsive. Reset type: SYSRSn
26	IPC26	R-0/W1S	0h	Writing 1 to this bit clears the IPC26 flag for the remote CPU. Writing 0 has no effect. Notes: [1] Normally, each CPU will clear (acknowledge) only its own local flags. This mechanism may be useful if the remote CPU is non-responsive. Reset type: SYSRSn
25	IPC25	R-0/W1S	0h	Writing 1 to this bit clears the IPC25 flag for the remote CPU. Writing 0 has no effect. Notes: [1] Normally, each CPU will clear (acknowledge) only its own local flags. This mechanism may be useful if the remote CPU is non-responsive. Reset type: SYSRSn
24	IPC24	R-0/W1S	0h	Writing 1 to this bit clears the IPC24 flag for the remote CPU. Writing 0 has no effect. Notes: [1] Normally, each CPU will clear (acknowledge) only its own local flags. This mechanism may be useful if the remote CPU is non-responsive. Reset type: SYSRSn
23	IPC23	R-0/W1S	0h	Writing 1 to this bit clears the IPC23 flag for the remote CPU. Writing 0 has no effect. Notes: [1] Normally, each CPU will clear (acknowledge) only its own local flags. This mechanism may be useful if the remote CPU is non-responsive. Reset type: SYSRSn
22	IPC22	R-0/W1S	0h	Writing 1 to this bit clears the IPC22 flag for the remote CPU. Writing 0 has no effect. Notes: [1] Normally, each CPU will clear (acknowledge) only its own local flags. This mechanism may be useful if the remote CPU is non-responsive. Reset type: SYSRSn
21	IPC21	R-0/W1S	0h	Writing 1 to this bit clears the IPC21 flag for the remote CPU. Writing 0 has no effect. Notes: [1] Normally, each CPU will clear (acknowledge) only its own local flags. This mechanism may be useful if the remote CPU is non-responsive. Reset type: SYSRSn
20	IPC20	R-0/W1S	0h	Writing 1 to this bit clears the IPC20 flag for the remote CPU. Writing 0 has no effect. Notes: [1] Normally, each CPU will clear (acknowledge) only its own local flags. This mechanism may be useful if the remote CPU is non-responsive. Reset type: SYSRSn

Table 7-28. IPCCLR Register Field Descriptions (continued)

Bit	Field	Type	Reset	Description
19	IPC19	R-0/W1S	0h	Writing 1 to this bit clears the IPC19 flag for the remote CPU. Writing 0 has no effect. Notes: [1] Normally, each CPU will clear (acknowledge) only its own local flags. This mechanism may be useful if the remote CPU is non-responsive. Reset type: SYSRSn
18	IPC18	R-0/W1S	0h	Writing 1 to this bit clears the IPC18 flag for the remote CPU. Writing 0 has no effect. Notes: [1] Normally, each CPU will clear (acknowledge) only its own local flags. This mechanism may be useful if the remote CPU is non-responsive. Reset type: SYSRSn
17	IPC17	R-0/W1S	0h	Writing 1 to this bit clears the IPC17 flag for the remote CPU. Writing 0 has no effect. Notes: [1] Normally, each CPU will clear (acknowledge) only its own local flags. This mechanism may be useful if the remote CPU is non-responsive. Reset type: SYSRSn
16	IPC16	R-0/W1S	0h	Writing 1 to this bit clears the IPC16 flag for the remote CPU. Writing 0 has no effect. Notes: [1] Normally, each CPU will clear (acknowledge) only its own local flags. This mechanism may be useful if the remote CPU is non-responsive. Reset type: SYSRSn
15	IPC15	R-0/W1S	0h	Writing 1 to this bit clears the IPC15 flag for the remote CPU. Writing 0 has no effect. Notes: [1] Normally, each CPU will clear (acknowledge) only its own local flags. This mechanism may be useful if the remote CPU is non-responsive. Reset type: SYSRSn
14	IPC14	R-0/W1S	0h	Writing 1 to this bit clears the IPC14 flag for the remote CPU. Writing 0 has no effect. Notes: [1] Normally, each CPU will clear (acknowledge) only its own local flags. This mechanism may be useful if the remote CPU is non-responsive. Reset type: SYSRSn
13	IPC13	R-0/W1S	0h	Writing 1 to this bit clears the IPC13 flag for the remote CPU. Writing 0 has no effect. Notes: [1] Normally, each CPU will clear (acknowledge) only its own local flags. This mechanism may be useful if the remote CPU is non-responsive. Reset type: SYSRSn
12	IPC12	R-0/W1S	0h	Writing 1 to this bit clears the IPC12 flag for the remote CPU. Writing 0 has no effect. Notes: [1] Normally, each CPU will clear (acknowledge) only its own local flags. This mechanism may be useful if the remote CPU is non-responsive. Reset type: SYSRSn

Table 7-28. IPCCLR Register Field Descriptions (continued)

Bit	Field	Type	Reset	Description
11	IPC11	R-0/W1S	0h	Writing 1 to this bit clears the IPC11 flag for the remote CPU. Writing 0 has no effect. Notes: [1] Normally, each CPU will clear (acknowledge) only its own local flags. This mechanism may be useful if the remote CPU is non-responsive. Reset type: SYSRSn
10	IPC10	R-0/W1S	0h	Writing 1 to this bit clears the IPC10 flag for the remote CPU. Writing 0 has no effect. Notes: [1] Normally, each CPU will clear (acknowledge) only its own local flags. This mechanism may be useful if the remote CPU is non-responsive. Reset type: SYSRSn
9	IPC9	R-0/W1S	0h	Writing 1 to this bit clears the IPC9 flag for the remote CPU. Writing 0 has no effect. Notes: [1] Normally, each CPU will clear (acknowledge) only its own local flags. This mechanism may be useful if the remote CPU is non-responsive. Reset type: SYSRSn
8	IPC8	R-0/W1S	0h	Writing 1 to this bit clears the IPC8 flag for the remote CPU. Writing 0 has no effect. Notes: [1] Normally, each CPU will clear (acknowledge) only its own local flags. This mechanism may be useful if the remote CPU is non-responsive. Reset type: SYSRSn
7	IPC7	R-0/W1S	0h	Writing 1 to this bit clears the IPC7 flag for the remote CPU. Writing 0 has no effect. Notes: [1] Normally, each CPU will clear (acknowledge) only its own local flags. This mechanism may be useful if the remote CPU is non-responsive. Reset type: SYSRSn
6	IPC6	R-0/W1S	0h	Writing 1 to this bit clears the IPC6 flag for the remote CPU. Writing 0 has no effect. Notes: [1] Normally, each CPU will clear (acknowledge) only its own local flags. This mechanism may be useful if the remote CPU is non-responsive. Reset type: SYSRSn
5	IPC5	R-0/W1S	0h	Writing 1 to this bit clears the IPC5 flag for the remote CPU. Writing 0 has no effect. Notes: [1] Normally, each CPU will clear (acknowledge) only its own local flags. This mechanism may be useful if the remote CPU is non-responsive. Reset type: SYSRSn
4	IPC4	R-0/W1S	0h	Writing 1 to this bit clears the IPC4 flag for the remote CPU. Writing 0 has no effect. Notes: [1] Normally, each CPU will clear (acknowledge) only its own local flags. This mechanism may be useful if the remote CPU is non-responsive. Reset type: SYSRSn

Table 7-28. IPCCLR Register Field Descriptions (continued)

Bit	Field	Type	Reset	Description
3	IPC3	R-0/W1S	0h	Writing 1 to this bit clears the IPC3 flag for the remote CPU. Writing 0 has no effect. Notes: [1] Normally, each CPU will clear (acknowledge) only its own local flags. This mechanism may be useful if the remote CPU is non-responsive. Reset type: SYSRSn
2	IPC2	R-0/W1S	0h	Writing 1 to this bit clears the IPC2 flag for the remote CPU. Writing 0 has no effect. Notes: [1] Normally, each CPU will clear (acknowledge) only its own local flags. This mechanism may be useful if the remote CPU is non-responsive. Reset type: SYSRSn
1	IPC1	R-0/W1S	0h	Writing 1 to this bit clears the IPC1 flag for the remote CPU. Writing 0 has no effect. Notes: [1] Normally, each CPU will clear (acknowledge) only its own local flags. This mechanism may be useful if the remote CPU is non-responsive. Reset type: SYSRSn
0	IPC0	R-0/W1S	0h	Writing 1 to this bit clears the IPC0 flag for the remote CPU. Writing 0 has no effect. Notes: [1] Normally, each CPU will clear (acknowledge) only its own local flags. This mechanism may be useful if the remote CPU is non-responsive. Reset type: SYSRSn

7.7.3.5 IPCFLG Register (Offset = 8h) [Reset = 00000000h]

IPCFLG is shown in Figure 7-23 and described in Table 7-29.

Return to the Summary Table.

IPC remote flag status register

Figure 7-23. IPCFLG Register

31	30	29	28	27	26	25	24
IPC31	IPC30	IPC29	IPC28	IPC27	IPC26	IPC25	IPC24
R-0h	R-0h	R-0h	R-0h	R-0h	R-0h	R-0h	R-0h
23	22	21	20	19	18	17	16
IPC23	IPC22	IPC21	IPC20	IPC19	IPC18	IPC17	IPC16
R-0h	R-0h	R-0h	R-0h	R-0h	R-0h	R-0h	R-0h
15	14	13	12	11	10	9	8
IPC15	IPC14	IPC13	IPC12	IPC11	IPC10	IPC9	IPC8
R-0h	R-0h	R-0h	R-0h	R-0h	R-0h	R-0h	R-0h
7	6	5	4	3	2	1	0
IPC7	IPC6	IPC5	IPC4	IPC3	IPC2	IPC1	IPC0
R-0h	R-0h	R-0h	R-0h	R-0h	R-0h	R-0h	R-0h

Table 7-29. IPCFLG Register Field Descriptions

Bit	Field	Type	Reset	Description
31	IPC31	R	0h	Indicates to the local CPU whether the remote IPC31 event flag is set. 0: The remote IPC31 event flag is not set 1: The remote IPC31 event flag is set Reset type: SYSRSn
30	IPC30	R	0h	Indicates to the local CPU whether the remote IPC30 event flag is set. 0: The remote IPC30 event flag is not set 1: The remote IPC30 event flag is set Reset type: SYSRSn
29	IPC29	R	0h	Indicates to the local CPU whether the remote IPC29 event flag is set. 0: The remote IPC29 event flag is not set 1: The remote IPC29 event flag is set Reset type: SYSRSn
28	IPC28	R	0h	Indicates to the local CPU whether the remote IPC28 event flag is set. 0: The remote IPC28 event flag is not set 1: The remote IPC28 event flag is set Reset type: SYSRSn
27	IPC27	R	0h	Indicates to the local CPU whether the remote IPC27 event flag is set. 0: The remote IPC27 event flag is not set 1: The remote IPC27 event flag is set Reset type: SYSRSn
26	IPC26	R	0h	Indicates to the local CPU whether the remote IPC26 event flag is set. 0: The remote IPC26 event flag is not set 1: The remote IPC26 event flag is set Reset type: SYSRSn

Table 7-29. IPCFLG Register Field Descriptions (continued)

Bit	Field	Type	Reset	Description
25	IPC25	R	0h	Indicates to the local CPU whether the remote IPC25 event flag is set. 0: The remote IPC25 event flag is not set 1: The remote IPC25 event flag is set Reset type: SYSRSn
24	IPC24	R	0h	Indicates to the local CPU whether the remote IPC24 event flag is set. 0: The remote IPC24 event flag is not set 1: The remote IPC24 event flag is set Reset type: SYSRSn
23	IPC23	R	0h	Indicates to the local CPU whether the remote IPC23 event flag is set. 0: The remote IPC23 event flag is not set 1: The remote IPC23 event flag is set Reset type: SYSRSn
22	IPC22	R	0h	Indicates to the local CPU whether the remote IPC22 event flag is set. 0: The remote IPC22 event flag is not set 1: The remote IPC22 event flag is set Reset type: SYSRSn
21	IPC21	R	0h	Indicates to the local CPU whether the remote IPC21 event flag is set. 0: The remote IPC21 event flag is not set 1: The remote IPC21 event flag is set Reset type: SYSRSn
20	IPC20	R	0h	Indicates to the local CPU whether the remote IPC20 event flag is set. 0: The remote IPC20 event flag is not set 1: The remote IPC20 event flag is set Reset type: SYSRSn
19	IPC19	R	0h	Indicates to the local CPU whether the remote IPC19 event flag is set. 0: The remote IPC19 event flag is not set 1: The remote IPC19 event flag is set Reset type: SYSRSn
18	IPC18	R	0h	Indicates to the local CPU whether the remote IPC18 event flag is set. 0: The remote IPC18 event flag is not set 1: The remote IPC18 event flag is set Reset type: SYSRSn
17	IPC17	R	0h	Indicates to the local CPU whether the remote IPC17 event flag is set. 0: The remote IPC17 event flag is not set 1: The remote IPC17 event flag is set Reset type: SYSRSn
16	IPC16	R	0h	Indicates to the local CPU whether the remote IPC16 event flag is set. 0: The remote IPC16 event flag is not set 1: The remote IPC16 event flag is set Reset type: SYSRSn
15	IPC15	R	0h	Indicates to the local CPU whether the remote IPC15 event flag is set. 0: The remote IPC15 event flag is not set 1: The remote IPC15 event flag is set Reset type: SYSRSn

Table 7-29. IPCFLG Register Field Descriptions (continued)

Bit	Field	Type	Reset	Description
14	IPC14	R	0h	Indicates to the local CPU whether the remote IPC14 event flag is set. 0: The remote IPC14 event flag is not set 1: The remote IPC14 event flag is set Reset type: SYSRSn
13	IPC13	R	0h	Indicates to the local CPU whether the remote IPC13 event flag is set. 0: The remote IPC13 event flag is not set 1: The remote IPC13 event flag is set Reset type: SYSRSn
12	IPC12	R	0h	Indicates to the local CPU whether the remote IPC12 event flag is set. 0: The remote IPC12 event flag is not set 1: The remote IPC12 event flag is set Reset type: SYSRSn
11	IPC11	R	0h	Indicates to the local CPU whether the remote IPC11 event flag is set. 0: The remote IPC11 event flag is not set 1: The remote IPC11 event flag is set Reset type: SYSRSn
10	IPC10	R	0h	Indicates to the local CPU whether the remote IPC10 event flag is set. 0: The remote IPC10 event flag is not set 1: The remote IPC10 event flag is set Reset type: SYSRSn
9	IPC9	R	0h	Indicates to the local CPU whether the remote IPC9 event flag is set. 0: The remote IPC9 event flag is not set 1: The remote IPC9 event flag is set Reset type: SYSRSn
8	IPC8	R	0h	Indicates to the local CPU whether the remote IPC8 event flag is set. 0: The remote IPC8 event flag is not set 1: The remote IPC8 event flag is set Reset type: SYSRSn
7	IPC7	R	0h	Indicates to the local CPU whether the remote IPC7 event flag is set. 0: The remote IPC7 event flag is not set 1: The remote IPC7 event flag is set Reset type: SYSRSn
6	IPC6	R	0h	Indicates to the local CPU whether the remote IPC6 event flag is set. 0: The remote IPC6 event flag is not set 1: The remote IPC6 event flag is set Reset type: SYSRSn
5	IPC5	R	0h	Indicates to the local CPU whether the remote IPC5 event flag is set. 0: The remote IPC5 event flag is not set 1: The remote IPC5 event flag is set Reset type: SYSRSn
4	IPC4	R	0h	Indicates to the local CPU whether the remote IPC4 event flag is set. 0: The remote IPC4 event flag is not set 1: The remote IPC4 event flag is set Reset type: SYSRSn
3	IPC3	R	0h	Indicates to the local CPU whether the remote IPC3 event flag is set. 0: The remote IPC3 event flag is not set 1: The remote IPC3 event flag is set Notes: [1] IPC event flags 0-3 will trigger interrupts in the receiving CPU via the ePIE. Reset type: SYSRSn

Table 7-29. IPCFLG Register Field Descriptions (continued)

Bit	Field	Type	Reset	Description
2	IPC2	R	0h	Indicates to the local CPU whether the remote IPC2 event flag is set. 0: The remote IPC2 event flag is not set 1: The remote IPC2 event flag is set Notes: [1] IPC event flags 0-3 will trigger interrupts in the receiving CPU via the ePIE. Reset type: SYSRSn
1	IPC1	R	0h	Indicates to the local CPU whether the remote IPC1 event flag is set. 0: The remote IPC1 event flag is not set 1: The remote IPC1 event flag is set Notes: [1] IPC event flags 0-3 will trigger interrupts in the receiving CPU via the ePIE. Reset type: SYSRSn
0	IPC0	R	0h	Indicates to the local CPU whether the remote IPC0 event flag is set. 0: The remote IPC0 event flag is not set 1: The remote IPC0 event flag is set Notes: [1] IPC event flags 0-3 will trigger interrupts in the receiving CPU via the ePIE. Reset type: SYSRSn

7.7.3.6 IPCCOUNTERL Register (Offset = Ch) [Reset = 00000000h]

IPCCOUNTERL is shown in Figure 7-24 and described in Table 7-30.

Return to the Summary Table.

IPC Counter Low Register

Figure 7-24. IPCCOUNTERL Register

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
COUNT																															
R-0h																															

Table 7-30. IPCCOUNTERL Register Field Descriptions

Bit	Field	Type	Reset	Description
31-0	COUNT	R	0h	This is the lower 32-bits of free running 64 bit timestamp counter clocked by the PLLSYSCLK. Reset type: CPU1.SYSRSn

7.7.3.7 IPCCOUNTERH Register (Offset = Eh) [Reset = 00000000h]

IPCCOUNTERH is shown in Figure 7-25 and described in Table 7-31.

Return to the Summary Table.

IPC Counter High Register

Figure 7-25. IPCCOUNTERH Register

31-0
COUNT
R-0h

Table 7-31. IPCCOUNTERH Register Field Descriptions

Bit	Field	Type	Reset	Description
31-0	COUNT	R	0h	This is the upper 32-bits of free running 64 bit timestamp counter clocked by the PLLSYSCLK. Reset type: CPU1.SYSRSn

7.7.3.8 IPCRECVCOM Register (Offset = 10h) [Reset = 00000000h]

IPCRECVCOM is shown in Figure 7-26 and described in Table 7-32.

Return to the Summary Table.

Remote to Local IPC Command Register

Figure 7-26. IPCRECVCOM Register

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
COMMAND																															
R-0h																															

Table 7-32. IPCRECVCOM Register Field Descriptions

Bit	Field	Type	Reset	Description
31-0	COMMAND	R	0h	This is a general purpose register used to receive software-defined commands from the remote CPU. It can only be written by the remote CPU. Notes [1] The local CPU's IPCRECVCOM is the same physical register as the remote CPU's IPCSENDCOM, and is located at the same address in both CPUs. [2] This register is reset by a SYRSn of the remote CPU Reset type: CPUx.SYSRSn

7.7.3.9 IPCRECVADDR Register (Offset = 12h) [Reset = 00000000h]

IPCRECVADDR is shown in Figure 7-27 and described in Table 7-33.

Return to the Summary Table.

Remote to Local IPC Address Register

Figure 7-27. IPCRECVADDR Register

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ADDRESS																															
R-0h																															

Table 7-33. IPCRECVADDR Register Field Descriptions

Bit	Field	Type	Reset	Description
31-0	ADDRESS	R	0h	This is a general purpose register used to receive software-defined addresses from the remote CPU. It can only be written by the remote CPU. Notes [1] The local CPU's IPCRECVADDR is the same physical register as the remote CPU's IPCSENDADDR, and is located at the same address in both CPUs. [2] This register is reset by a SYRSn of the remote CPU Reset type: CPUx.SYSRSn

7.7.3.10 IPCRECVDATA Register (Offset = 14h) [Reset = 00000000h]

IPCRECVDATA is shown in Figure 7-28 and described in Table 7-34.

Return to the Summary Table.

Remote to Local IPC Data Register

Figure 7-28. IPCRECVDATA Register

31-0
WDATA
R-0h

Table 7-34. IPCRECVDATA Register Field Descriptions

Bit	Field	Type	Reset	Description
31-0	WDATA	R	0h	This is a general purpose register used to receive software-defined data from the remote CPU. It can only be written by the remote CPU. Notes [1] The local CPU's IPCRECVDATA is the same physical register as the remote CPU's IPCSENDDATA, and is located at the same address in both CPUs. [2] This register is reset by a SYRSn of the remote CPU Reset type: CPUx.SYSRSn

7.7.3.11 IPCLOCALREPLY Register (Offset = 16h) [Reset = 00000000h]

IPCLOCALREPLY is shown in Figure 7-29 and described in Table 7-35.

Return to the Summary Table.

Local to Remote IPC Reply Data Register

Figure 7-29. IPCLOCALREPLY Register

31-0
RDATA
R/W-0h

Table 7-35. IPCLOCALREPLY Register Field Descriptions

Bit	Field	Type	Reset	Description
31-0	RDATA	R/W	0h	This is a general purpose register used to send software-defined data to the remote CPU in response to a command. It can only be written by the local CPU. Notes [1] The local CPU's IPCLOCALREPLY is the same physical register as the remote CPU's IPCREMOTEREPLY, and is located at the same address in both CPUs. Reset type: SYSRSn

7.7.3.12 IPCSENDCOM Register (Offset = 18h) [Reset = 00000000h]

IPCSENDCOM is shown in Figure 7-30 and described in Table 7-36.

Return to the Summary Table.

Local to Remote IPC Command Register

Figure 7-30. IPCSENDCOM Register

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
COMMAND																															
R/W-0h																															

Table 7-36. IPCSENDCOM Register Field Descriptions

Bit	Field	Type	Reset	Description
31-0	COMMAND	R/W	0h	This is a general purpose register used to send software-defined commands to the remote CPU. It can only be written by the local CPU. Notes [1] The local CPU's IPCSENDCOM is the same physical register as the remote CPU's IPCRECVCOM, and is located at the same address in both CPUs. Reset type: SYSRSn

7.7.3.13 IPCSENDADDR Register (Offset = 1Ah) [Reset = 00000000h]

IPCSENDADDR is shown in Figure 7-31 and described in Table 7-37.

Return to the Summary Table.

Local to Remote IPC Address Register

Figure 7-31. IPCSENDADDR Register

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ADDRESS																															
R/W-0h																															

Table 7-37. IPCSENDADDR Register Field Descriptions

Bit	Field	Type	Reset	Description
31-0	ADDRESS	R/W	0h	This is a general purpose register used to send software-defined addresses to the remote CPU. It can only be written by the local CPU. Notes [1] The local CPU's IPCSENDADDR is the same physical register as the remote CPU's IPCRECVDATA, and is located at the same address in both CPUs. Reset type: SYSRSn

7.7.3.14 IPCSENDDATA Register (Offset = 1Ch) [Reset = 00000000h]

IPCSENDDATA is shown in Figure 7-32 and described in Table 7-38.

Return to the Summary Table.

Local to Remote IPC Data Register

Figure 7-32. IPCSENDDATA Register

31–0
WDATA
R/W-0h

Table 7-38. IPCSENDDATA Register Field Descriptions

Bit	Field	Type	Reset	Description
31-0	WDATA	R/W	0h	This is a general purpose register used to send software-defined data to the remote CPU. It can only be written by the local CPU. Notes [1] The local CPU's IPCSENDDATA is the same physical register as the remote CPU's IPCRECVDATA, and is located at the same address in both CPUs. Reset type: SYSRSn

7.7.3.15 IPCREMOTEREPLY Register (Offset = 1Eh) [Reset = 00000000h]

IPCREMOTEREPLY is shown in Figure 7-33 and described in Table 7-39.

Return to the Summary Table.

Remote to Local IPC Reply Data Register

Figure 7-33. IPCREMOTEREPLY Register

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
RDATA																															
R-0h																															

Table 7-39. IPCREMOTEREPLY Register Field Descriptions

Bit	Field	Type	Reset	Description
31-0	RDATA	R	0h	This is a general purpose register used to receive software-defined data from the remote CPU's response to a command. It can only be written by the remote CPU. Notes [1] The local CPU's IPCREMOTEREPLY is the same physical register as the remote CPU's IPCLOCALREPLY, and is located at the same address in both CPUs. [2] This register is reset by a SYRSn of the remote CPU Reset type: CPUx.SYSRSn

7.7.3.16 IPCBOOTSTS Register (Offset = 20h) [Reset = 00000000h]

IPCBOOTSTS is shown in Figure 7-34 and described in Table 7-40.

Return to the Summary Table.

CPU2 to CPU1 IPC Boot Status Register

Figure 7-34. IPCBOOTSTS Register

31-0
BOOTSTS
R/W-0h

Table 7-40. IPCBOOTSTS Register Field Descriptions

Bit	Field	Type	Reset	Description
31-0	BOOTSTS	R/W	0h	This register is used by CPU2 to pass the boot Status to CPU1. The data format is software-defined. It can only be written by CPU2. Reset type: CPU2.SYSRSn

7.7.3.17 IPCBOOTMODE Register (Offset = 22h) [Reset = 00000000h]

IPCBOOTMODE is shown in Figure 7-35 and described in Table 7-41.

Return to the Summary Table.

CPU1 to CPU2 IPC Boot Mode Register

Figure 7-35. IPCBOOTMODE Register

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
BOOTMODE																															
R/W-0h																															

Table 7-41. IPCBOOTMODE Register Field Descriptions

Bit	Field	Type	Reset	Description
31-0	BOOTMODE	R/W	0h	This register is used by CPU1 to pass a boot mode information to CPU2. The data format is software-defined. It can only be written by CPU1. Reset type: CPU1.SYSRSn

7.7.4 IPC Registers to Driverlib Functions

Table 7-42. IPC Registers to Driverlib Functions

File	Driverlib Function
ACK	
-	
STS	
-	
SET	
-	
CLR	
-	
FLG	
-	
COUNTERL	
-	
COUNTERH	
-	
SENDCOM	
-	
SENDADDR	
-	
SENDDATA	
-	
REMOTEREPLY	
-	
RECVCOM	
-	
RECVADDR	
-	
RECVDATA	
-	

Table 7-42. IPC Registers to Driverlib Functions (continued)

File	Driverlib Function
LOCALREPLY	
-	
RECVCOM	
-	
RECVADDR	
-	
RECVDATA	
-	
LOCALREPLY	
-	
SENDCOM	
-	
SENDADDR	
-	
SENDDATA	
-	
REMOTEREPLY	
-	
BOOTSTS	
-	
BOOTMODE	
-	